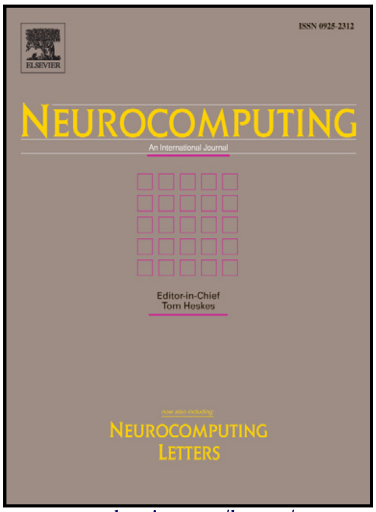

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Sparse Kernel Entropy Component Analysis for Dimensionality Reduction of Biomedical Data

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Abstract

Dimensionality reduction is ubiquitous in biomedical applications. A newly proposed spectral dimensionality reduction method, named Kernel entropy component analysis (KECA), can reveal the structure related to Renyi entropy of an input space data set. However, each principal component in the Hilbert space depends on all training samples in KECA, causing degraded performance. To overcome this drawback, a sparse KECA (SKECA) algorithm based on a recursive divide-and-conquer (DC) method is proposed in this work. The original large and complex problem of KECA is decomposed into a series of small and simple sub-problems, and then they are solved recursively. The performance of SKECA is evaluated on four common biomedical datasets, and compared with KECA, principal component analysis (PCA), kernel PCA (KPCA), sparse PCA and sparse KPCA. Experimental results indicate that the SKECA outperforms conventional dimensionality reduction algorithms, even for high order dimensional features. It suggests that SKECA is potentially applicable to biomedical data processing.

Keywords: Sparse kernel entropy component analysis; Divide-and-conquer method; Dimensionality reduction; Biomedical data

1. Introduction

Machine learning has been widely applied in the domain of biomedicine to perform computer-aided detection, diagnosis and analysis in the past decade. Biomedical data are usually characterized by high dimensionality but with relatively small instances. Consequently, curse of dimensionality becomes a major issue in biomedical data processing [1]. Since much of the information in high-dimensional data is concentrated in lower-dimensional spaces, dimensionality reduction has become important in biomedical data processing [1][2].

Principal component analysis (PCA) and its nonlinear extension, Kernel PCA (KPCA) are the most classical dimensionality reduction methods, which have been widely used for biomedical data. Although there are many variants of PCA and KPCA [3][4][5], they are still typical spectral data transformation methods, which realize dimensionality reduction by selecting the top eigenvalues (spectra) and corresponding eigenvectors of the specially constructed data matrixes [6]. However, the transformation may be based on uninformative eigenvectors from the viewpoint of information theory [7].

Kernel entropy component analysis (KECA) is a newly developed information-theory-based data transformation method. This method preserves the most estimated Renyi entropy in the input space data set via a kernel-based estimator [8]. Different from other spectral methods, in KECA, the selection of top eigenvalues and eigenvectors of the kernel matrix is not necessary, and data transformation reveals the intrinsic structure related to information entropy of the input space data [8]. Moreover, the KECA-transformed data typically has a distinct angular structure, implying that each of these eigenvectors carries information about the cluster structure [8]. With these properties, KECA has been successfully applied to feature extraction and clustering, and shown improved performance over PCA and KPCA [7][13].

Like PCA and KPCA, however, KECA is computationally expensive when processing large-scale samples, because each principal component in the Hilbert space of KECA depends on all training samples, contradicting the goal of obtaining informative and concise representations [8]. In fact, not all original variables are physically meaningful or have contribution to the final principal components [11]. For example, in gene expression data, each variable corresponds to a certain gene, therefore, if the derived principal components are sparse, they can facilitate their interpretability. Thus, one natural approach to overcome this limitation is integrating the sparse solution into KECA algorithms, because sparsity can reduce the computation complexity, improve the component relevance and interpretability, and better reveal the data's underlying structure [14]. For PCA and KPCA, sparse PCA (SPCA) and sparse KPCA (SKPCA) algorithms have been proposed in recent years [14]-[20]. But, to the best of our knowledge, there is no sparse KECA (SKECA) algorithm that has been proposed.

On the other hand, the recursive divide-and-conquer (DC) method has been recently used in machine learning [21]-[24]. It recursively breaks down a problem into sub-problems of the same (or related) type until they become simple enough to be solved directly. Solutions to the sub-problems are then combined to give solution to the original problem [21]-[24]. Using this idea, DC-based SPCA decomposes the original large and complex PCA problem into a series of small and simple sub-problems, and then recursively solve them [18]. Experimental results have indicated the effectiveness of SPCA.

In this work, a DC-based sparse KECA (SKECA) method is proposed motivated by DC solution due to its simplicity and effectiveness [18]. Since many biomedical data in general have small samples but with high dimensional features and should be processed with effective dimensionality reduction methods, the proposed SKECA is then applied to reduce the feature dimensions of biomedical data.

The rest of the paper is organized as follows. Section 2 provides an introduction to KECA and the proposed DC-based SKECA algorithm. Experiments and results are given in Section 3 to evaluate the performance of SKECA, with discussions presented in Section 4.

2. Sparse KECA Algorithm

2.1 Brief review of KECA

The original KECA algorithm [8] is briefly introduced as follows. The Renyi quadratic entropy is defined as

$$H(p) = -\log \int p^2(\mathbf{x})d\mathbf{x} \tag{1}$$

where $p(\mathbf{x})$ is a probability density function to generate the dataset $S = \mathbf{x}_1,\ldots,\mathbf{x}_N$. Due to the monotonic property of logarithmic function, Eq. (1) can be focused on the quantity $V(p) = \int p^2(\mathbf{x})d\mathbf{x}$, which denotes the expectation of $V(p)$ w.r.t. the density $p(\mathbf{x})$. Then, the Parzen window density estimator is employed to estimate $V(p)$ by [25]

$$\hat{p}(x) = \frac{1}{N}\sum_{\mathbf{x}_t \in S} \mathbf{k}_\sigma(\mathbf{x}, \mathbf{x}_t) \tag{2}$$

where $\mathbf{k}_\sigma(\mathbf{x},\mathbf{x}_t)$ is the Parzen window or kernel centered at $\mathbf{x}_t$, and σ the width parameter. By using the sample mean approximation of the expectation operator, the following equation can be obtained

$$\begin{aligned}\hat{V}(p) &= \frac{1}{N}\sum_{\mathbf{x}_t \in S}\hat{p}(\mathbf{x}_t) = \frac{1}{N}\sum_{\mathbf{x}_t \in S}\frac{1}{N}\sum_{\mathbf{x}_{t'} \in S}\mathbf{k}_\sigma(\mathbf{x}, \mathbf{x}_{t'}) \\ &= \frac{1}{N^2}\mathbf{1}^{\mathrm{T}}\mathbf{K}\mathbf{1}\end{aligned} \tag{3}$$

where the element (t, t') of the $N \times N$ kernel matrix $\mathbf{K}$ equals to $k_\sigma(\mathbf{x}_t, \mathbf{x}_{t'})$, and $\mathbf{1}$ is an ($N\times1$) vector of ones. Thus, the empirical Renyi entropy estimation resides in the elements of the corresponding kernel matrix [26].

The Renyi entropy estimator then can be expressed based on the eigenvalues and eigenvectors of the kernel matrix, which decompose $\mathbf{K}$ as $\mathbf{K}=\mathbf{E}\mathbf{D}\mathbf{E}^{\mathrm{T}}$, where $\mathbf{D}$ is a diagonal matrix containing eigenvalues λ_1, …, λ_N, and $\mathbf{E}$ is a matrix with the corresponding eigenvectors $\mathbf{e}_1,\ldots,\mathbf{e}_N$ as its columns. Eq. (3) can be rewritten as

$$\hat{V}(p) = \frac{1}{N^2}\sum_{i=1}^{N}(\sqrt{\lambda_i}\mathbf{e}_i^{\mathrm{T}}\mathbf{1})^2 \tag{4}$$

As can be seen from Eq. (4) that some of the eigenvalues and their corresponding eigenvectors contribute more to the entropy estimates than others, as the terms depend on different eigenvalues and eigenvectors.

Let: $R^d \to F$ denote a nonlinear map, then $\mathbf{x}_t \to \boldsymbol{\varphi}(\mathbf{x}_t)$. Let $\boldsymbol{\Phi}(\mathbf{x})=[\boldsymbol{\varphi}(\mathbf{x}_1),\ldots,\boldsymbol{\varphi}(\mathbf{x}_N)]$. The inner-product (kernel) matrix $\mathbf{K}=\boldsymbol{\Phi}^{\mathrm{T}}\boldsymbol{\Phi}$ is then obtained. A projection of $\boldsymbol{\Phi}$ onto a single principal axis is given by $\mathbf{u}_i^{\mathrm{T}}\boldsymbol{\Phi}=\lambda_i^{1/2}\mathbf{e}_i^{\mathrm{T}}$. Thus, projections of $\boldsymbol{\Phi}$ onto all principal axes are given by

$$\mathbf{U}^{\mathrm{T}}\boldsymbol{\Phi}=\mathbf{D}^{1/2}\mathbf{E}^{\mathrm{T}} \tag{5}$$

where $\mathbf{U}=[\mathbf{u}_1,\ldots,\mathbf{u}_N]$ is the projection matrix.

KECA is defined as a k-dimensional data transformation by projecting $\boldsymbol{\Phi}$ onto a subspace $\mathbf{U}_k$ spanned by those k feature space principal axes, which contribute most to the Renyi entropy estimate of data. The transformation is denoted by

$$\boldsymbol{\Phi}_{\mathrm{eca}} = \mathbf{U}_k^{\mathrm{T}}\boldsymbol{\Phi} = \mathbf{D}_k^{1/2}\mathbf{E}_k^{\mathrm{T}} \tag{6}$$

where $\mathbf{D}_k \in R^{k\times k}$ is a diagonal matrix containing selected k eigenvalues, and $\mathbf{E}_k \in R^{N\times k}$ is a matrix with the corresponding k eigenvectors as columns. The solution of Eq. (6) is obtained by minimizing the following:

$$\mathbf{\Phi}_{\text{eca}} = \mathbf{D}_k^{1/2}\mathbf{E}_k^{\mathrm{T}} : \min_{\lambda_1,\mathbf{e}_1,\ldots,\lambda_N,\mathbf{e}_N} \hat{V}(p) - \hat{V}_k(p) \tag{7}$$

where $\hat{V}_k(p) = \frac{1}{N^2}\sum_{i=1}^{k}(\sqrt{\lambda_i}\mathbf{e}_i^{\mathrm{T}}\mathbf{1})^2$. Furthermore, for out-of-sample (test) data point $\boldsymbol{\varphi}(\mathbf{x})$, one can obtain an explicit expression for projections onto a principal axis $\mathbf{u}_i$ in the kernel space F. Requiring unitary norm $\|\mathbf{u}_i\|^2=1$, we thus have $\mathbf{u}_i=\lambda^{-1/2}\mathbf{\Phi}\mathbf{e}_i$. Therefore

$$\mathbf{u}_i^{\mathrm{T}}\boldsymbol{\varphi}(\mathbf{x}) = \left\langle \lambda_i^{-1/2}\sum_{t=1}^{N}\mathbf{e}_{i,t}\boldsymbol{\varphi}(\mathbf{x}_t), \boldsymbol{\varphi}(\mathbf{x}) \right\rangle = \lambda_i^{-1/2}\sum_{t=1}^{N}\mathbf{e}_{i,t}K(\mathbf{x}_t,\mathbf{x}) \tag{8}$$

where $\mathbf{e}_{i,t}$ denotes the t-th element of $\mathbf{e}_i$. Let $\mathbf{\Phi}^*$ be a collection of out-of-sample data points, and let the inner-product matrix $\mathbf{K}^*=\mathbf{\Phi}^{\mathrm{T}}\mathbf{\Phi}^*$. Then, by Eq. (8), we obtain

$$\mathbf{\Phi}_{\text{eca}}' = \mathbf{U}_k^{\mathrm{T}}\mathbf{\Phi}^* = \mathbf{D}_k^{-1/2}\mathbf{E}_k^{\mathrm{T}}\mathbf{\Phi}^{\mathrm{T}}\mathbf{\Phi}^* = \mathbf{D}_k^{-1/2}\mathbf{E}_k^{\mathrm{T}}\mathbf{K}^* \tag{9}$$

Likewise, for training date sets **Φ,** Eq. (6) can be re-written as

$$\mathbf{\Phi}_{\text{eca}} = \mathbf{D}_k^{-1/2}\mathbf{E}_k^{\mathrm{T}}\mathbf{\Phi}^{\mathrm{T}}\mathbf{\Phi} = \mathbf{D}_k^{-1/2}\mathbf{E}_k^{\mathrm{T}}\mathbf{K} \tag{10}$$

Thus, data transformation is achieved by projecting $\mathbf{\Phi}$ onto the axes that contribute most to the Renyi entropy of the input space data. Therefore, KECA is different from the top eigenvalues based KPCA.

2.2 Sparse KECA

The final transformation formula of PCA only has two terms, namely, projection matrix and input data matrix. Most of the SPCA algorithms only need the sparse projection matrix that can select better features by sparsity [14]-[17]. However, the final transformation formula of KECA has three terms as seen from Eq. (10). The first term $\mathbf{D}_k^{-1/2}$ is a diagonal matrix, and the third term $\mathbf{K}$ is a kernel matrix, whose elements correspond to the Renyi entropy estimation [26]. Since not all Renyi entropy estimates are useful, our SKECA focuses on achieving sparsity from the second term $\mathbf{E}_k^{\mathrm{T}}$, which can force the useless or redundant Renyi entropy estimates to zero.

To describe the SKECA algorithm more clearly, we divide the process of SKECA into two stages, namely sparsity with DC solution and data transformation.

1) Sparsity with DC solution

It is worth noting that in Eq. (7), $\hat{V}_k(p)$ preserves the Renyi entropy estimate $\hat{V}(p)$ of the original data $\mathbf{x}_1, \ldots, \mathbf{x}_N$ as much as possible from an input space perspective. Like kernel matrix $\mathbf{K}=\mathbf{E}\mathbf{D}\mathbf{E}^{\mathrm{T}}$, we define

$$\mathbf{K}_{\text{eca}} = \mathbf{E}_k \mathbf{D}_k \mathbf{E}_k^{\mathrm{T}} \tag{11}$$

$\mathbf{K}_{\text{eca}}$ is associated with a data transformation such that the input and the transformed data entropies are most similar. Thus, Eq. (7) can be rewritten as following by combining Eq. (3) and (4):

$$\begin{aligned}\mathbf{\Phi}_{\text{eca}} = \mathbf{D}_k^{1/2}\mathbf{E}_k^{\mathrm{T}} : & \min_{\lambda_1,\mathbf{e}_1,\ldots,\lambda_N,\mathbf{e}_N} \hat{V}(p) - \hat{V}_k(p) \\ = & \min_{\lambda_1,\mathbf{e}_1,\ldots,\lambda_N,\mathbf{e}_N} \frac{1}{N^2}\mathbf{1}^{\mathrm{T}}(\mathbf{K}-\mathbf{K}_{eca})\mathbf{1}\end{aligned} \tag{12}$$

Obviously, minimization of Eq. (12) mainly depends on ($\mathbf{K}$-$\mathbf{K}_{\text{eca}}$). Therefore, a solution to the sparse problem of SKECA can be obtained based on Eq. (12):

$$\begin{aligned}\min_{\mathbf{K}_{\text{eca}}} \left\|\mathbf{K}-\mathbf{K}_{\text{eca}}\right\|_F^2 &= \min_{\mathbf{E}_k,\mathbf{D}_k} \left\|\mathbf{K}-\mathbf{E}_k\mathbf{D}_k\mathbf{E}_k^{\mathrm{T}}\right\|_F^2 \\ &= \min_{\mathbf{E}_k,\mathbf{D}_k} \left\|\mathbf{K}-\mathbf{U}\mathbf{V}^{\mathrm{T}}\right\|_F^2 \\ \text{s.t. } \mathbf{v}_i^{\mathrm{T}}\mathbf{v}_i &= 1, \left\|\mathbf{v}_i\right\|_p \le t_i \, (i=1,\ldots,k)\end{aligned} \tag{13}$$

where $\mathbf{U}=\mathbf{E}_k\mathbf{D}_k \in R^{N\times k}$, $\mathbf{V}=\mathbf{E}_k \in R^{N\times k}$, and k is defined as k-dimension in SKECA. p = 0 or 1 in $\left\|\mathbf{v}_i\right\|_p$ denotes l_0-norm or l_1-norm of $\mathbf{v}_i$, which enforces sparsity of the output sparse $\mathbf{v}_i$ [15][17]. t_i is a sparse parameter to control the number of nonzero elements in $\mathbf{V}$ corresponding to the final sparse entropy component. The object function of Eq. (13) is then formulated as:

$$\left\|\mathbf{K}-\mathbf{U}\mathbf{V}\right\|_F^2 = \left\|\mathbf{K}-\sum_{j=1}^{k}\mathbf{u}_j\mathbf{v}_j^{\mathrm{T}}\right\|_F^2 = \left\|\mathbf{E}_i-\mathbf{u}_i\mathbf{v}_i^{\mathrm{T}}\right\|_F^2 \tag{14}$$

where $\mathbf{E}_i = \mathbf{K} - \sum_{j \neq i} \mathbf{u}_j \mathbf{v}_j^{\mathrm{T}}$. Therefore, according to the idea of DC method [18], it is easy to separate the original large minimization problem into a series of small ones, each with respect to a column vector $\mathbf{u}_i$ of $\mathbf{U}$ and $\mathbf{v}_i$ of $\mathbf{V}$ for i=1,…, k, respectively, which is as follows:

$$\min_{\mathbf{v}_i} \left\| \mathbf{E}_i - \mathbf{u}_i \mathbf{v}_i^{\mathrm{T}} \right\|_F^2 \ \text{ s.t. } \ \mathbf{v}_i^{\mathrm{T}} \mathbf{v}_i = 1, \left\| \mathbf{v}_i \right\|_p \le t_i \, (i = 1,...,k) \tag{15}$$

and

$$\min_{\mathbf{u}_i} \left\| \mathbf{E}_i - \mathbf{u}_i \mathbf{v}_i^{\mathrm{T}} \right\|_F^2 \tag{16}$$

For the convenience of denotation, we rewrite Eqs. (15) and (16) as the following:

$$\min_{\mathbf{v}} \left\| \mathbf{E} - \mathbf{u}\mathbf{v}^{\mathrm{T}} \right\|_F^2 \ \text{ s.t. } \ \mathbf{v}^{\mathrm{T}} \mathbf{v} = 1, \ \left\| \mathbf{v} \right\|_p \le t \tag{17}$$

and

$$\min_{\mathbf{u}} \left\| \mathbf{E} - \mathbf{u}\mathbf{v}^{\mathrm{T}} \right\|_F^2 \tag{18}$$

Then, Eqs. (17) and (18) are equivalent to the following optimization problems [18], respectively:

$$\max_{\mathbf{v}} (\mathbf{E}^{\mathrm{T}} \mathbf{u})^{\mathrm{T}} \mathbf{v} \ \text{ s.t. } \ \mathbf{v}^{\mathrm{T}} \mathbf{v} = 1, \ \left\| \mathbf{v} \right\|_p \le t \tag{19}$$

and

$$\min_{\mathbf{u}} \mathbf{u}^{\mathrm{T}} \mathbf{u} - 2(\mathbf{E}\mathbf{v})^{\mathrm{T}} \mathbf{u} \tag{20}$$

The closed-form solutions of Eqs. (19) and (20) can then be solved by the algorithm in [18].

When p =0 of Eq. (19), the optimal solution of Eq. (19) is given by:

$$\mathbf{v}_0^*(\mathbf{w}, t) = \begin{cases} \phi, & t < 1 \\ \dfrac{hard_{\theta_k}(\mathbf{w})}{\left\| hard_{\theta_k}(\mathbf{w}) \right\|_2}, & k \le t < k+1 \ (k = 1, 2, ...d-1) \\ \dfrac{\mathbf{w}}{\left\| \mathbf{w} \right\|_2} & t \ge d \end{cases} \tag{21}$$

where $\mathbf{w}$=$\mathbf{E}^{\mathrm{T}}\mathbf{u}$, θ_k denotes the k-th largest element of $|\mathbf{w}|$ and ϕ denotes the empty set, and $hard_{\theta_k}(\mathbf{w})$

is the hard thresholding function [18].

In the case of p = 1, Eq. (19) has the following closed-form solution. We first denote

$$f_{\mathbf{w}}(\lambda) = \frac{soft_{\lambda}(\mathbf{w})}{\left\| soft_{\lambda}(\mathbf{w}) \right\|_2} \tag{22}$$

where $soft_{\lambda}(\mathbf{w})$ is the soft thresholding function [18]. Then the optimal solution of Eq. (19) is given by

$$\mathbf{v}_0^{*}(\mathbf{w}, t) = \begin{cases} \phi, & t < 1 \\ f_{\mathbf{w}}(\lambda_k), & \left\| f_{\mathbf{w}}(\left|\mathbf{w}_{I_k}\right|) \right\|_1 \le t < \left\| f_{\mathbf{w}}(\left|\mathbf{w}_{I_{k-1}}\right|) \right\|_1 (k = 2, 3, .., d-1) \\ f_{\mathbf{w}}(\lambda_1), & \left\| f_{\mathbf{w}}(\left|\mathbf{w}_{I_1}\right|) \right\|_1 \le t < \sqrt{d} \\ f_{\mathbf{w}}(0), & t \ge \sqrt{d} \end{cases} \tag{23}$$

After calculating **v**, the optimal solution of Eq. (20) is directly given by **u=Ev** [18].

Therefore, the DC solution can be summarized to calculate **U** and **V** to recursively optimize each column $\mathbf{u}_i$ of **U** and $\mathbf{v}_i$ of **V** with another $\mathbf{u}_j$ or $\mathbf{v}_j$ fixed, respectively [18]. By doing it iteratively, **U** and **V** are recursively updated until a stopping criterion is satisfied. The stopping criterion is that the change of **U** or **V** is less than a preset threshold, or a maximum number of iterations are reached. The DC solution and complete description of SKECA algorithm are summarized in Fig. 1.

2) Data transformation

In the first stage of SKECA, sparsity is achieved from the second term $\mathbf{E}_k^{\mathrm{T}}$ of Eq. (10). Therefore, we define SKECA transformation as a k-dimensional data transformation with sparse matrix **V** taking the place of $\mathbf{E}_k^{\mathrm{T}}$ in Eq. (10). We have

$$\mathbf{\Phi}_{\text{skeca}} = \mathbf{U}_k^{\mathrm{T}}\mathbf{\Phi} = \mathbf{D}_k^{1/2}\mathbf{V}^{\mathrm{T}} = \mathbf{D}_k^{-1/2}\mathbf{V}^{\mathrm{T}}\mathbf{K} \tag{24}$$

where $\mathbf{U}_k = [\mathbf{u}_1, \ldots, \mathbf{u}_k]$, $\mathbf{u}_i^{\mathrm{T}} = \lambda_i^{1/2}\mathbf{v}_i^{\mathrm{T}}$, $\mathbf{V} = [\mathbf{v}_1, \ldots, \mathbf{v}_k] \in R^{N\times k}$. **V** is obtained in the first stage. λ_i and **K** are the same as those defined in the original KECA.

For the out-of-sample data point $\boldsymbol{\varphi}(\mathbf{x})$, like KECA, the corresponding mapped feature of SKECA is obtained by

$$\mathbf{\Phi}_{\text{skeca}}{}' = \mathbf{U}_k{}^{\mathrm{T}}\ \mathbf{\Phi}^* = \mathbf{D}_k{}^{-1/2}\mathbf{V}^{\mathrm{T}}\mathbf{K}^* \tag{25}$$

where $\mathbf{K}^*$is the same as in Eq. (9).

2.3 Sparse KPCA

Based on the DC method [18], we also propose a SKPCA algorithm with a similar solution to the SKECA for the purpose of comparison (see the next section). The main difference between SKECA and SKPCA is in the initializations of **U** and **V**. Although the formulations of initializations of **U** and **V** are the same ($\mathbf{U}=\mathbf{E}_k\mathbf{D}_k$, and $\mathbf{V}=\mathbf{E}_k$), $\mathbf{D}_k$ and $\mathbf{E}_k$ are different as they depend on the selection of different eigenvalues and eigenvectors in the SKECA or SKPCA.

3. Experiments and Results

To demonstrate benefits of the proposed SKECA algorithm, comparative experiments were performed on four commonly used datasets for evaluating machine learning methods.

SKECA was compared with the PCA, KPCA, original KECA, SPCA [18], and the SKPCA described in section 2.3. The commonly used k-nearest neighbor (KNN) algorithm was selected as the classifier, because compared with other classifiers with strong learning ability, such as support vector machine, the classification results more depend on the feature other than KNN. Four typical biomedical datasets were used in experiment to indicate feasibility of dimensionality reduction by SKECA for biomedical data. The 5-fold cross-validation strategy was performed for the first three datasets, and the 4-fold cross-validation strategy for breast ultrasound image dataset. Classification accuracy was used as an evaluation index. Sensitivity and specificity were also selected to evaluate the performance of different algorithms for breast ultrasound image classification. The results are represented as mean±SD. We also evaluate the effect of sparsity on classification accuracy with two experiments: 1) only the i-th component in the sparse matrix **V** is sparse, and 2) all the first i components in the sparse matrix $\mathbf{V}$ are sparse.

3.1 Results on SMK-CAN-187 Dataset

The first experiment was conducted to evaluate dimensionality reduction performance of SKECA for high dimensional features, because biomedical data usually have high dimensional features but small samples. The SMK-CAN-187 Dataset, a microarray gene expression dataset from the ASU Feature Selection Repository [27], is tested here. This dataset contains 2 classes and 187 instances with 19993 features. Reduced features from 20 to 100 dimensions with an interval of 20 were selected in this study. The mean classification accuracy of the original 19993 features is 73.30±5.99%.

Table 1 shows classification results of different algorithms with different reduced feature dimensions. As can be seen that SKECA achieves the best performance for each given feature dimension among all other compared algorithms, which indicates that SKECA can effectively reduce dimensionality of high dimensional data. The best classification accuracy of SKECA is 83.97±1.75% with 80 dimensional features, which achieves improvements up to 10.1% compared with the best accuracy of KECA, and improves at least 4.8%, compared with all other algorithms. Note that all sparsity-based algorithms outperform the corresponding non-sparse algorithms.

Since this gene expression dataset has the typical small sample problem but with very high feature dimension, we evaluate the running time of SKECA on this dataset as an example. Table 2 shows the running times of different algorithms when the feature dimension is reduced to 20-dimension. It can be found in Table 2 that SKECA has the similar running time to KECA, KPCA and SKPCA, and all these algorithms are much faster than PCA and SPCA, which indicates the efficiency of SKECA.

Fig. 2 shows the sparse effect of only i-th (i=1, 2, 3, 4, 5) component in the sparse matrix **V** on classification accuracy for SKECA, SKPCA and SPCA. Here all classification accuracies are the best at its corresponding i-th component by adjusting the value of sparse parameter t_i. All the three sparsity-based algorithms achieve best accuracy at the first sparse component. Increasing i usually results in worse performance. Fig. 3 shows the sparse effect of all the first i (i=1, 2, 3, 4, 5) components on classification accuracy with two groups of sparse parameters (t=[1, 1, 1, 1, 1] and t=[5, 5, 5, 5, 5]) for SKECA, SKPCA and SPCA. Here, each value in t is the number of nonzero elements of the corresponding sparse component. As seen from Fig. 3, although performance of sparsity-based algorithms can be improved with increased sparse components, more sparse components do not necessarily result in better performance.

3.2 Results on Sleep Apnea Dataset

We then evaluate the performance of SKECA on the dataset with more samples. The Sleep Apnea Dataset [28] contains 25 full overnight polysomnograms with simultaneous three-channel Holter ECG, from 25 adult subjects with suspected sleep-disordered breathing. It is a multi-class problem with five sleep stages. The EEG channel (C3-A2) signal from two subjects is randomly selected. All signals were preprocessed by notch filtering at 50 Hz to suppress power line disturbances, and then filtered with a band-pass filter from 0.3 to 64 Hz. All EEG signals were then segmented with the epoch length of 30s without overlap. Each epoch before and after a sleep stage transition was removed to avoid possible subsections of mislabeled data within one epoch [29]. There are a total of 935 instances for testing with 176, 253, 112, 216 and 178 instances corresponding to the S1, S2, SWS, REM and awake stages, respectively. A total of 78 features were extracted by calculating means, standard deviations, kurtosis, entropy, and other coefficients extracted from fast Fourier transform and wavelet transform. The mean classification accuracy of the original 78 features was 71.01±2.22%.

As shown in Table 3, SKECA outperforms all other algorithms almost at each feature dimension only at 10-dimension, which again indicate the effectiveness of SKECA for dimensionality reduction. The best classification accuracy obtained by SKECA is 75.51±3.31% with 50 dimensional features, and improves 4.1% compared with the best classification accuracy KECA, which shows that SKECA is more efficient, and can better handle more samples than original KECA due to the sparse effect in SKECA.

Fig. 4 shows the sparse effect of only the i-th component in the sparse matrix $\mathbf{V}$ on classification accuracy for SKECA, SKPCA and SPCA. Again, all algorithms achieve best accuracy at the first sparse component, and the classification performance degrades with increased i. Fig. 5 shows the sparse effect of all the first i components on classification accuracy for SKECA, SKPCA and SPCA. Also, performance is not monotonically improved with the increased number of sparse components.

3.3 Results on 2D Hela Cell Image Dataset

Medical images are the very common data in the biomedical field, therefore, the third experiment was conducted on the Hela Cell Image Dataset, which provides the most commonly used grayscale images in machine learning [30]**Error! Reference source not found.**. This dataset includes fluorescence microscopy images of HeLa cells stained with various organelle-specific fluorescent dyes. It includes 10 organelles, and therefore is a ten-class problem. This database contains 862 images. Since texture feature is one of the most used feature extraction method for medical images, in this study, the uniform local binary patterns (u-LBP) algorithm was used to extract texture feature due to its effectivness for Hela Dataset [32]. The bin with the higher occurrence in u-LBP is discarded because it represents the black [32]. The LBP feature dimension is 243. We tested reduced features from 20 to 140 dimensions with an interval of 20. The mean classification accuracy of original 243 features is 86.08±1.44%.

Table 4 shows the classification results of different algorithms with different reduced feature dimensions. Similar to the results on Table 1 and 2, SKECA again outperforms other algorithms at each feature dimension. SKECA achieves the best accuracy of 88.28±0.77% with 140 dimensional features, and improves 2.4% compared with the best accuracy of KECA due to sparse effect. In this case, performance of PCA, KPCA and KECA is no better than the original 243 features. However, SKECA still improve the classification results, which indicates its effectiveness for dimensionality reduction.

Figs. 6 and 7 show the sparse effect of only the i-th component sparsity and all the first i components on classification accuracy for SKECA, SKPCA and SPCA, respectively. The curve trends are similar to those from Figs. 2 - 5, indicating that the first sparse component usually achieves the best performance, and more sparse components do not always give better results.

3.4 Results on Breast Ultrasound Image Dataset

The final experiment was conducted on the most commonly used clinical imaging data. The breast ultrasound image dataset was acquired from the Cancer Hospital of Fudan University in our previous work [33]. There are 200 pathology-proved breast ultrasound images in total (100 benign masses and 100 malignant tumors). The shearlet-based texture features were extracted, which were shown effective in our previous work [33]. The dimension of shearlet-based texture features is 148. We reduced the dimension to 20-140 with an interval of 20. For the original 148 features, its mean classification accuracy, mean sensitivity and mean specificity are 91.00±4.16%, 93.87±7.27% and 88.76±4.03%, respectively.

Tables 5, 6 and 7 show classification accuracies, sensitivities and specificities of different algorithms with different reduced feature dimensions, respectively. Both in Table 4 and 7, SKECA still outperforms other algorithms at each feature dimension on classification accuracy and specificity, respectively. Compared with KECA, SKECA provides improvements up 2.0%, 1.670% and 3.1% on classification accuracy, sensitivity and specific, respectively.

Figure 8 and 9 show the sparse effect of only the *i*-th component sparsity and all the first *i* components on classification accuracy for SKECA, SKPCA and SPCA, respectively. Once again, the results are similar to those shown in Figs. 2-7 on other datasets.

4. Discussion

In this paper, a DC-based SKECA algorithm is proposed to overcome the drawback of the original KECA, that all principal components in the Hilbert space depend on every training sample. The proposed method is then applied to reduce the feature dimensionality of biomedical data. Four biomedical datasets are used to evaluate the dimensionality reduction performance of SKECA. The experimental results indicate that SKECA has superior over original KECA and other conventional algorithms in terms of classification accuracy. Even for very high dimensional features in microarray gene expression dataset, SKECA also performances very well. It is worth noting that the datasets used in this study are very typical in biomedical research, including physiological signal, gene data and medical images. SKECA has shown its capability of dealing with various types of data as mentioned above, and also the best performance in dimensionality reduction among all the tested methods, implying that SKECA is potentially applicable to biomedical data.

From all the tables in Section 3, sparse-based algorithms usually outperform non-sparse algorithms. This is because the sparse-based algorithms do not use all original data variables to produce the final principal components. In biomedical data, the original variables have meaningful physical interpretations. For example, each variable of gene expression data corresponds to a certain gene. Therefore, the derived principal component loadings are always expected to be sparse so as to facilitate interpretability. Moreover, the original data inevitably contain redundant or useless elements, which are used in non-sparse KECA, KPCA and PCA, resulting in degraded performance. Sparse-based KECA, KPCA and PCA can effectively improve data transformation performance as compared with their non-sparse counterparts. These sparse-based approaches are particularly suitable for dimensionality reduction of biomedical data. As expected, SKECA performs better than SKPCA and SPCA because SKECA still maintains the property of original KECA, whose data transformation reveals structure related to the Renyi entropy of the input space data.

We also find that the efficiency/running time of SKECA is similar to other algorithms that are compared, namely KECA, KPCA and SKPCA, and they are all much faster than PCA and SPCA. For these kernel based dimensionality reduction algorithms, the larger the size of the training samples, the more the running time and the lower the efficiency of feature extraction, because it will calculate all the kernel functions between a new sample and the total training samples for further feature extraction with these functions. While for PCA and SPCA, the higher the feature dimensions, the more the running time, because the size of covariance matrix in PCA depends on the feature dimension. Therefore, SKECA is also competitive on running efficiency for biomedical data, because the biomedical data usually have small samples, but can provide very high dimensional features.

In this work, we have also studied the effect of sparsity on classification accuracy. The first sparse component in most cases leads to the best performance because it contains major information in KECA, KPCA and PCA. A sparsity operation on the first component makes it more effective with less redundancy. As can be seen from Figure 3, 5, 7 and 9, the number of sparse components is not necessarily monotonically related to the performance of sparse-based algorithms because more principal components do not always mean more useful information. The above results suggest that the first component should be sparse in SKECA, while the choice of a suitable number of sparse components still depends on experience. Therefore, our study in the future will be focused on finding a systematic approach of determining the number of sparse components in SKECA.

Acknowledgments

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References

[1] Y. Saeys, I. N. Inza, P. Larrañaga, A review of feature selection techniques in bioinformatics, Bioinformatics 23(19) (2007) 2507-2517.

[2] Q.H. Huang, D.C. Tao, X.L. Li, L.W. Jin, Exploiting local coherent patterns for unsupervised feature ranking, IEEE Transactions on Systems, Man and Cybernetics Part B-Cybernetics 41(6) (2011) 1471-1482.

[3] E. Candes, X.D. Li, Y. Ma, J. Wright, Robust principal component analysis, Journal of the ACM, 58(3) (2011) 11.

[4] X.L. Li, Y.W. Pang, Y. Yuan, L1-Norm-Based 2D PCA, IEEE Transactions on Systems, Man, and Cybernetics, Part B 40(4) (2010) 1170-1175.

[5] T,J. Chin, D. Suter, Incremental kernel principal component analysis, IEEE Transactions on Image Processing, 16(6) 2007 1662-1674.

[6] L. Xiao, J. Sun, S. Boyd, A duality view of spectral methods for dimensionality reduction, In: Proceedings of the 23rd International Conference on Machine Learning (ICML), 2006, pp. 1041-1048.

[7] Z.H. Zhang, E.R. Hancock, Kernel entropy-based unsupervised spectral feature slection, International Journal of Pattern Recognition and Artificial Intelligence 26(5) (2012) 1260002.

[8] R. Jenssen, Kernel entropy component analysis, IEEE Transactions on Pattern Analysis and Machine Intelligence 32(5) (2010) 847-860.

[9] B.H. Shekar, M.S. Kumari, L.M. Mestetskiy, N.F. Dyshkant, Face recognition using kernel entropy component analysis, Neurocomputing 74(6) (2011) 1053-1057.

[10] L. Gómez-Chova, R. Jenssen, G. Camps-Valls, Kernel entropy component analysis for remote sensing image clustering, IEEE Geoscience and Remote Sensing Letters 9(2) (2012) 312-316.

[11] Z.B. Xie, L. Guan, Multimodal information fusion of audio emotion recognition based on kernel entropy component analysis, Semantic Computing 7(1) (2013) 25-42.

[12] R. Jenssen, Entropy-relevant dimensions in kernel feature space, IEEE Signal Processing Magazine 30(4) (2013) 30-39.

[13] J. Shi, Q.K. Jiang, R. Mao, M.H. Lu, T.F. Wang. FR-KECA: Fuzzy Robust Kernel Entropy Component Analysis. Neurocomputing. 149 (2015) 1415-1423.

[14] M. Grbovic, C. R. Dance, S. Vucetic, Sparse principal component analysis with constraints, In: Proceedings of 26th AAAI Conference on Artificial Intelligence, 2012, pp. 935-941.

[15] H. Zou, T. Hastie, R. Tibshirani, Sparse principal component analysis, Journal of Computational and Graphical Statistics, 15(2) (2006) 265-286.

[16] F.R. Bach, L.E. Ghaoui, A. Aspremont, Full regularization path for sparse principal component analysis, In: Proceedings of 24th International Conference on Machine Learning (ICML), pp. 177-184, 2007.

[17] H.P. Shen, J.H. Z. Huang, Sparse principal component analysis via regularized low rank matrix approximation, Journal of Multivariate Analysis, 99(6) (2008) 1015-1034.

[18] Q. Zhao, D.Y. Meng, Z.B. Xu, A recursive divide-and-conquer approach for sparse principal component analysis, arXiv: 1211.7219, 2013.

[19] M.E. Tipping, Sparse kernel principal component analysis, In: Advances in Neural Information Processing Systems 13 (NIPS), vol. 1, pp. 211-244, 2001.

[20] J.B. Li, H.J. Gao, Sparse data-dependent kernel principal component analysis based on least squares support vector machine for feature extraction and recognition, Neural Computing & Applications, 21(8) (2012) 1971-1980.

[21] X.C. Wang, X.L. Wang, D.M. Wilkes, A divide-and-conquer approach for minimum spanning tree-based clustering, IEEE Transactions on Knowledge and Data Engineering, 21(7) (2009) 945-958.

[22] I. Blanes, J. Serra-Sagristà, M.W. Marcellin, J. Bartrina-Rapesta, Divide-and-conquer strategies for hyperspectral image processing: a review of their benefits and advantages, IEEE Signal Processing Magazine, 29(3) (2012) 71-81.

[23] E.S. Coakley, V. Rokhlin, A fast divide-and-conquer algorithm for computing the spectra of real symmetric tridiagonal matrices, Applied and Computational Harmonic Analysis, 34(3) (2013) 379-414.

[24] C.J. Hsieh, S. Si, I.S. Dhillon, A divide-and-conquer solver for kernel support vector machines, In: Proceedings of 31th International Conference on Machine Learning (ICML), 2014, pp. 566-574.

[25] E. Parzen, On the estimation of a probability density function and the mode, The Annals of Math. Statistics, vol. 33(3) (1962) 1065-1076.

[26] M. Girolami, Orthogonal series density estimation and the kernel eigenvalue problem, Neural Computation , 14(3) (2002) 669-688.

[27] http://featureselection.asu.edu/

[28] A.L. Goldberger, L.A. Amaral, L. Glass et al., Physiobank, physiotoolkit, and physionet: components of a new research resource for complex physiologic signals, Circulation, 101(23) (2000) 215-220.

[29] M. Längkvist, L. Karlsson, A. Loutfi, Sleep stage classification using unsupervised feature learning, Advances in Artificial Neural Systems, 2012(5) 2012, 107046.

[30] M. Boland, R.F. Murphy, A neural network classifier capable of recognizing the patterns of all major sub-cellular structures in fluorescence microscope images of heLa cells, Bioinformatics, 17(12) (2001) 1213-1223.

[31] http://murphylab.web.cmu.edu/data/

[32] L. Nanni, A. Lumini, S. Brahnam, Survey on LBP based texture descriptors for image classification, Expert Systems with Applications, 39(3) (2012) 3634-3641.

[33] S. Zhou, J. Shi, J. Zhu, et al. Shearlet-based texture feature extraction for classification of breast tumor in ultrasound image, Biomedical Signal Processing and Control, 8(6) (2013) 688-696.

Figure Captions

Figure 1 SKECA algorithm

Figure 2 The sparse effect of only the *i*-th component on classification accuracy on the SMK-CAN-187 Dataset.

Figure 3 The sparse effect of the first *i* components on classification accuracy on the SMK-CAN-187 Dataset. (a) SKECA; (b) SKPCA; (c) SPCA.

Figure 4 The sparse effect of only the *i*-th component on classification accuracy on the Sleep Apnea Dataset.

Figure 5 The sparse effect of all the first *i* components on classification accuracy on the Sleep Apnea Dataset. (a) SKECA; (b) SKPCA; (c) SPCA.

Figure 6 The sparse effect of only the *i*-th component on classification accuracy on the 2D Hela Cell Image Dataset.

Figure 7 The sparse effect of all the first *i* components on classification accuracy on the 2D Hela Cell Image Dataset. (a) SKECA; (b) SKPCA; (c) SPCA.

Figure 8 The sparse effect of only the *i*-th component on classification accuracy on the breast ultrasound image dataset.

Figure 9 The sparse effect of all the first *i* components on classification accuracy on the breast ultrasound image dataset. (a) SKECA; (b) SKPCA; (c) SPCA.

Table Captions

Table 1 Classification results of different algorithms with different reduced feature dimensions on SMK-CAN-187 Dataset (Unit: %)

Table 2 Running times of different algorithms on SMK-CAN-187 Dataset with 20-dimensional feature (Unit: second)

Table 3 Classification results of different algorithms with different reduced feature dimensions on Sleep Apnea Dataset (Unit: %)

Table 4 Classification results of different algorithms with different reduced feature dimensions on 2D Hela Cell Image Dataset (Unit: %)

Table 5 Accuracies of different algorithms with different reduced feature dimensions on Breast Ultrasound Image Dataset (Unit: %)

Table 6 Sensitivities of different algorithms with different reduced feature dimensions on Breast Ultrasound Image Dataset (Unit: %)

Table 7 Specificities of different algorithms with different reduced feature dimensions on Breast Ultrasound Image Dataset (Unit: %)

Table 1 Classification results of different algorithms with different reduced feature dimensions on SMK-CAN-187 Dataset (Unit: %)

Algorithm \ **Dimension**	**20**	**40**	**60**	**80**	**100**
PCA	71.71±8.29	72.22±6.53	72.76±5.52	73.32±5.92	71.72±9.40
KPCA	74.37±5.28	73.30±6.57	73.84±5.51	72.79±5.99	72.25±5.79
KECA	73.30±4.69	72.75±4.98	71.17±6.62	73.29±4.78	73.83±4.89
SPCA	74.88±3.88	75.41±2.16	75.95±1.68	78.08±1.02	77.03±2.07
SKPCA	76.50±4.91	77.58±7.72	79.19±6.69	77.58±6.98	76.53±6.61
SKECA	**82.94±5.91**	**81.31±3.09**	**81.85±3.76**	***83.97±1.75***	**81.32±4.78**

Table 2 Running times of different algorithms on SMK-CAN-187 Dataset with 20-dimensional features (Unit: second)

Algorithm	**PCA**	**KPCA**	**KECA**	**SPCA**	**SKPCA**	**SKECA**
Running Time	9.28±0.88	0.11±0.01	0.10±0.01	11.59±1.12	0.15±0.02	0.14±0.01

Table 3 Classification results of different algorithms with different reduced feature dimensions on Sleep Apnea Dataset (Unit: %)

Dimension / Algorithm	10	20	30	40	50	60	70
PCA	69.84±3.22	68.88±2.12	70.27±2.56	70.80±2.64	70.91±2.29	71.12±2.36	71.01±2.22
KPCA	69.41±4.10	68.77±2.79	70.37±2.58	70.91±2.67	71.23±2.44	71.12±2.36	71.01±2.22
KECA	56.79±5.16	67.06±5.91	69.41±1.98	71.44±2.44	71.23±2.44	71.12±2.36	71.01±2.22
SPCA	69.95±3.22	70.16±2.96	70.59±2.70	71.12±2.78	72.41±2.44	71.34±2.60	71.23±2.49
SKPCA	**70.69±3.73**	70.48±2.68	71.44±2.69	71.98±2.72	72.73±2.54	72.09±2.31	71.98±2.22
SKECA	64.92±2.13	**72.62±2.71**	**74.22±3.59**	**74.33±3.21**	***75.51±3.31***	**74.87±3.36**	**74.76±3.58**

Table 4 Classification results of different algorithms with different reduced feature dimensions on 2D Hela Cell Image Dataset (Unit: %)

Dimension / Algorithm	20	40	60	80	100	120	140
PCA	85.38±3.14	85.27±2.22	85.50±3.03	84.92±2.85	85.03±2.36	85.26±2.52	84.10±2.84
KPCA	85.50±3.15	85.27±2.22	85.50±3.03	84.92±2.85	85.14±2.42	85.26±2.52	84.10±2.84
KECA	85.38±2.23	85.38±1.78	85.04±2.37	85.61±1.51	85.61±1.51	85.50±1.61	85.84±1.95
SPCA	86.20±1.86	86.31±1.87	86.31±1.78	86.77±2.37	86.43±2.16	86.55±2.90	86.43±2.65
SKPCA	85.97±2.82	86.43±2.47	86.89±3.08	85.73±3.06	86.19±2.82	86.19±3.20	85.84±2.69
SKECA	**87.59±1.86**	**87.24±1.26**	**87.24±1.17**	**87.70±0.92**	**87.58±0.74**	**87.93±1.11**	***88.28±0.77***

Table 5 Accuracies of different algorithms with different reduced feature dimensions on Breast

Ultrasound Image Dataset (Unit: %)

Dimension / Algorithm	20	40	60	80	100	120	140
PCA	90.50±2.52	90.00±4.32	90.50±4.73	90.50±3.42	89.50±3.79	90.50±5.00	91.00±4.16
KPCA	90.50±2.52	90.50±5.00	90.50±4.73	91.00±3.83	90.50±4.73	91.00±5.29	91.50±4.43
KECA	91.00±3.46	91.00±3.46	90.50±4.73	92.00±2.83	92.50±3.42	91.00±4.16	91.00±4.16
SPCA	91.00±2.58	92.00±4.32	92.50±5.00	92.00±5.16	92.00±5.16	92.50±4.43	93.50±3.42
SKPCA	91.50±3.00	92.50±3.42	93.00±4.76	92.50±5.51	92.50±5.51	92.50±5.51	93.00±4.16
SKECA	**92.50±4.73**	**93.50±4.43**	**93.50±4.43**	**93.50±4.43**	***94.50±3.42***	***94.50±3.42***	**94.00±3.65**

Table 6 Sensitivities of different algorithms with different reduced feature dimensions on Breast Ultrasound Image Dataset (Unit: %)

Dimension / Algorithm	20	40	60	80	100	120	140
PCA	91.74±6.01	94.52±7.86	92.68±8.75	91.87±7.58	90.56±5.99	93.87±7.27	93.87±7.27
KPCA	93.67±7.33	93.68±9.45	93.04±8.89	90.56±5.99	90.56±5.99	93.87±7.27	93.87±7.27
KECA	93.71±5.72	94.83±7.90	93.56±7.17	94.52±7.86	94.83±7.90	93.87±7.27	93.87±7.27
SPCA	96.24±3.15	**97.04±3.82**	96.50±4.73	**96.50±4.73**	**96.50±4.73**	**96.50±4.73**	**95.67±6.29**
SKPCA	**97.33±3.27**	96.67±6.67	**96.67±6.67**	95.67±6.29	95.67±6.29	94.83±7.90	**95.67±6.29**
SKECA	95.54±4.13	96.50±4.73	96.50±4.73	**96.50±4.73**	95.83±8.33	95.83±8.33	94.83±7.90

Table 7 Specificities of different algorithms with different reduced feature dimensions on Breast Ultrasound Image Dataset (Unit: %)

Dimension / Algorithm	20	40	60	80	100	120	140
PCA	90.21±3.42	85.71±4.38	87.96±2.92	90.76±2.62	87.90±2.95	86.71±2.83	88.76±4.03
KPCA	88.12±6.64	88.00±5.38	89.00±3.41	91.80±1.46	90.55±3.79	88.55±4.76	89.80±4.13
KECA	88.71±3.09	87.91±3.22	87.51±4.28	90.01±5.16	91.01±3.82	88.76±4.03	88.76±4.03
SPCA	85.21±6.91	88.90±5.65	88.07±6.86	87.07±7.16	87.07±7.16	88.32±5.83	91.82±4.76
SKPCA	85.46±4.46	89.00±3.41	89.55±3.81	89.36±6.00	89.36±6.00	90.61±5.27	90.57±4.28
SKECA	**89.35±7.05**	**90.36±5.07**	**90.36±5.07**	**90.36±5.07**	**93.99±5.52**	**93.99±5.52**	**94.11±5.33**

Highlights

1. We propose a sparse kernel entropy component analysis (SKECA) algorithm.
2. The SKECA algorithm is based on the divide-and-conquer method.
3. The SKECA algorithm is applied to the dimensionality reduction of biomedical data.
4. Experimental results show that the SKECA outperforms other conventional algorithms.

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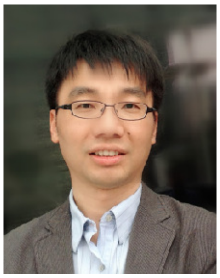
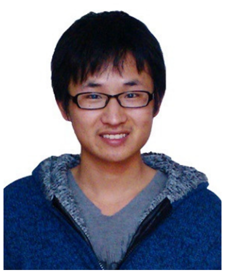

Jun Shi


Qikun Jiang


Qi Zhang


Qinghua Huang

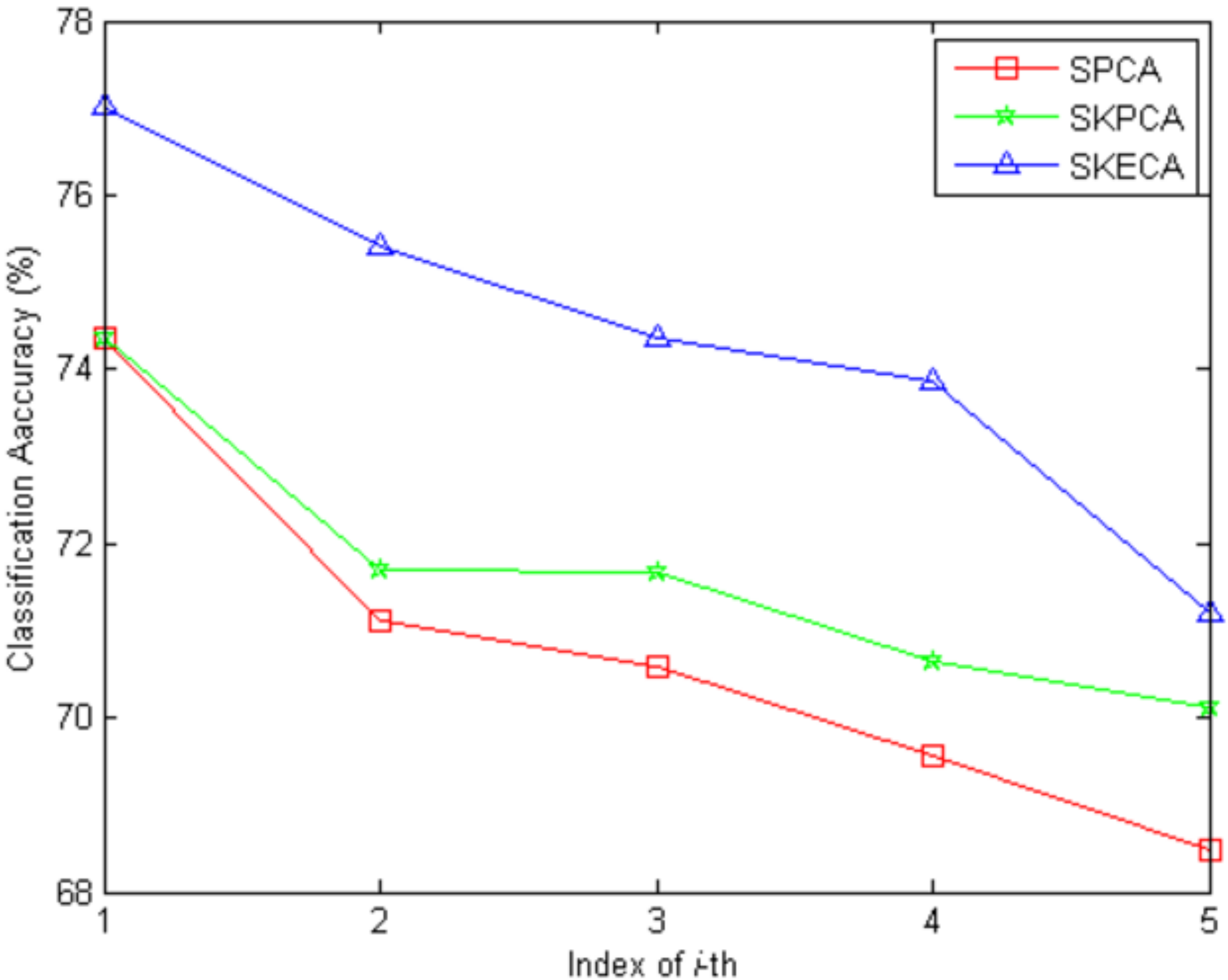
(Photo unavailable) Xuelong Li

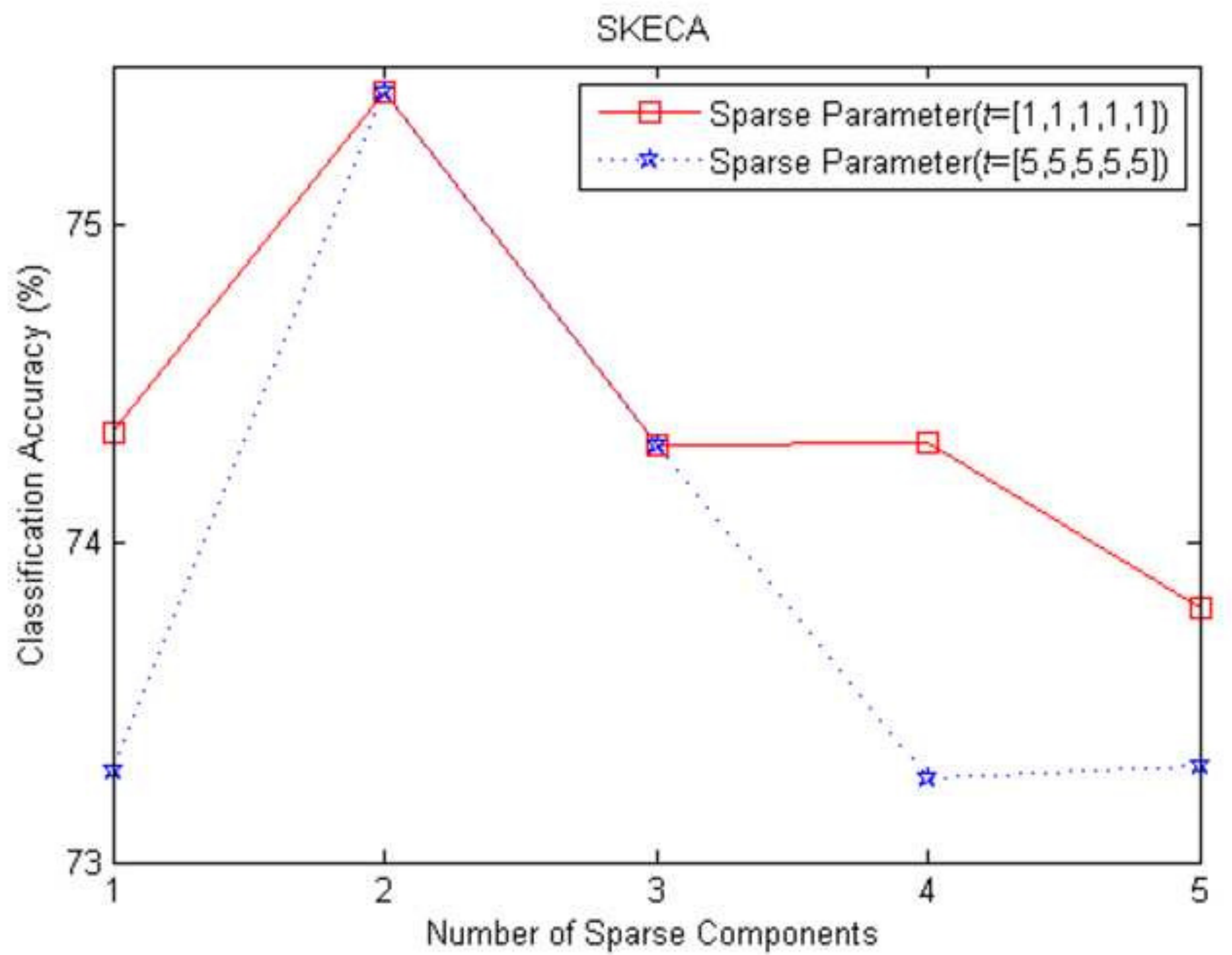
Input: Data matrix $\mathbf{X} \in R^{d \times N}$, the number of kernel entropy principle entropy component k, necessary parameters in DC method: the type of constraint norm p (p=0 or p=1), the maximum number of iterations, the thresholding for stop and sparse parameter $t = (t_1, \ldots, t_k)$.

1. Calculate kernel matrix $\mathbf{K}$, and then decompose $\mathbf{K}$ as $\mathbf{K}=\mathbf{EDE}^{\mathrm{T}}$.
2. Calculate $\mathbf{E}_k$ and $\mathbf{D}_k$, Initialize $\mathbf{U}=\mathbf{E}_k\mathbf{D}_k \in R^{N \times k}$ and $\mathbf{V}= \mathbf{E}_k \in R^{N \times k}$.
3. **Repeat**
4. **For** i=1,…,k, do
5. Calculate $\mathbf{E}_i = \mathbf{K} - \sum_{j \neq i} \mathbf{u}_j \mathbf{v}_j^{\mathrm{T}}$.
6. Update $\mathbf{v}_i$ via solving (15) based on DC method.
7. Update $\mathbf{u}_i$ via solving (16) based on DC method.
8. **End for**
9. **Until** stopping criterion satisfied.

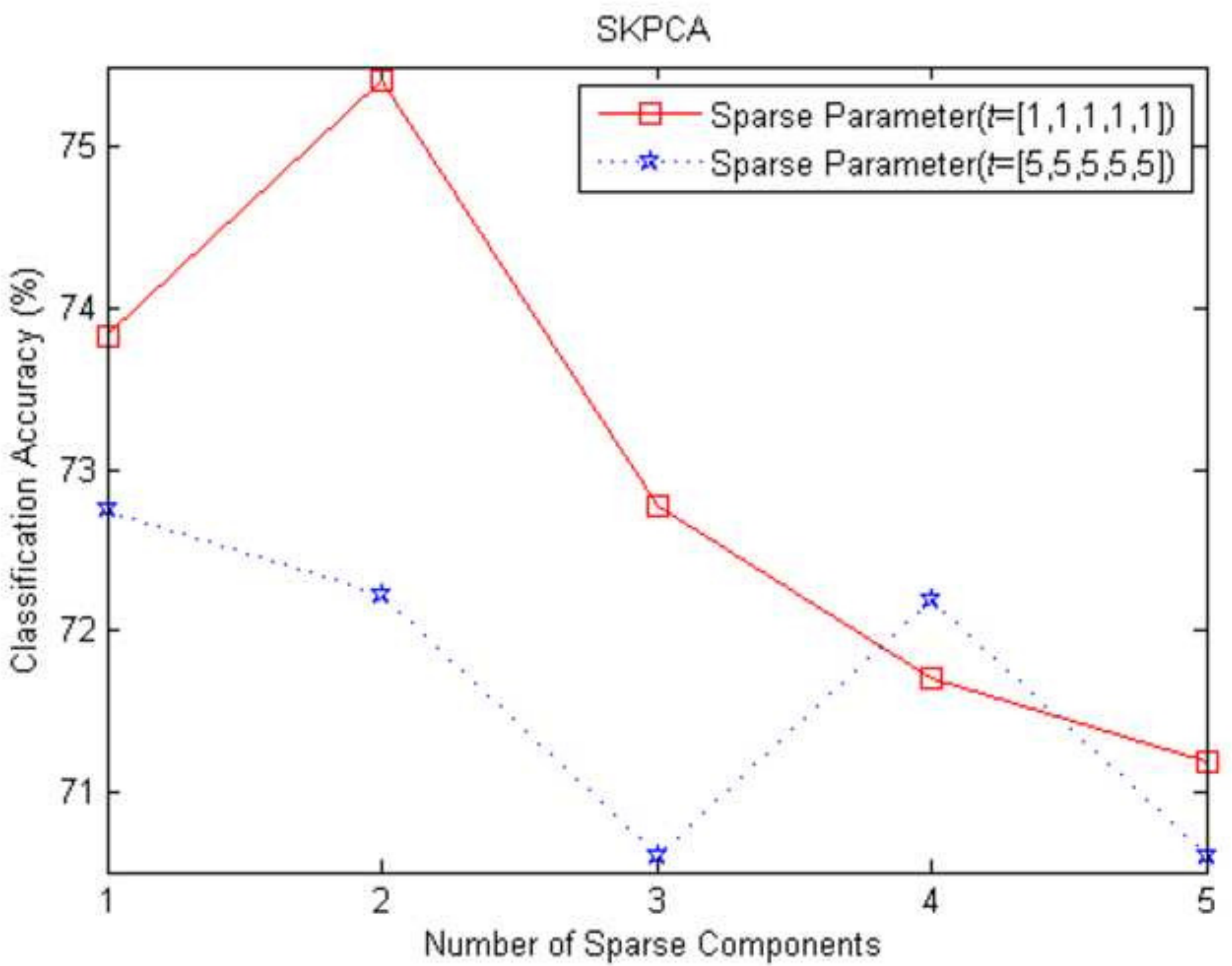
Output: The transformation result of SKECA algorithm:

$$\mathbf{\Phi}_{\mathrm{skeca}} = \mathbf{U}_k^{\mathrm{T}} \mathbf{\Phi} = \mathbf{D}_k^{1/2} \mathbf{V}^{\mathrm{T}} = \mathbf{D}_k^{-1/2} \mathbf{V}^{\mathrm{T}} \mathbf{K}$$

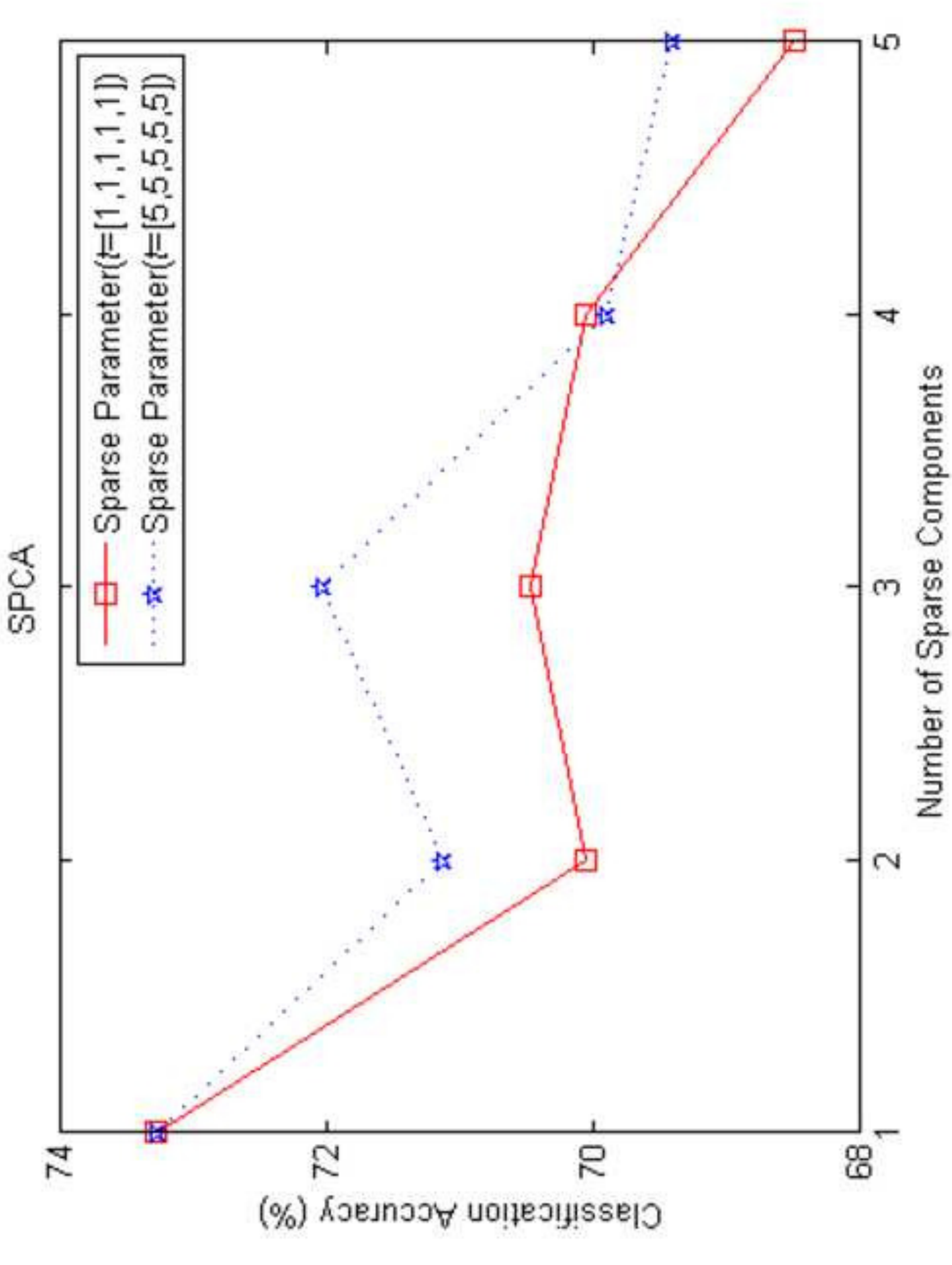

SPCA
SKPCA
SKECA
Classification Aaccuracy (%)
Index of i-th
78
76
74
72
70
68
1
2
3
4
5


SKECA
Sparse Parameter(t=[1,1,1,1,1])
Sparse Parameter(t=[5,5,5,5,5])
Classification Accuracy (%)
75
74
73
1
2
3
4
5
Number of Sparse Components

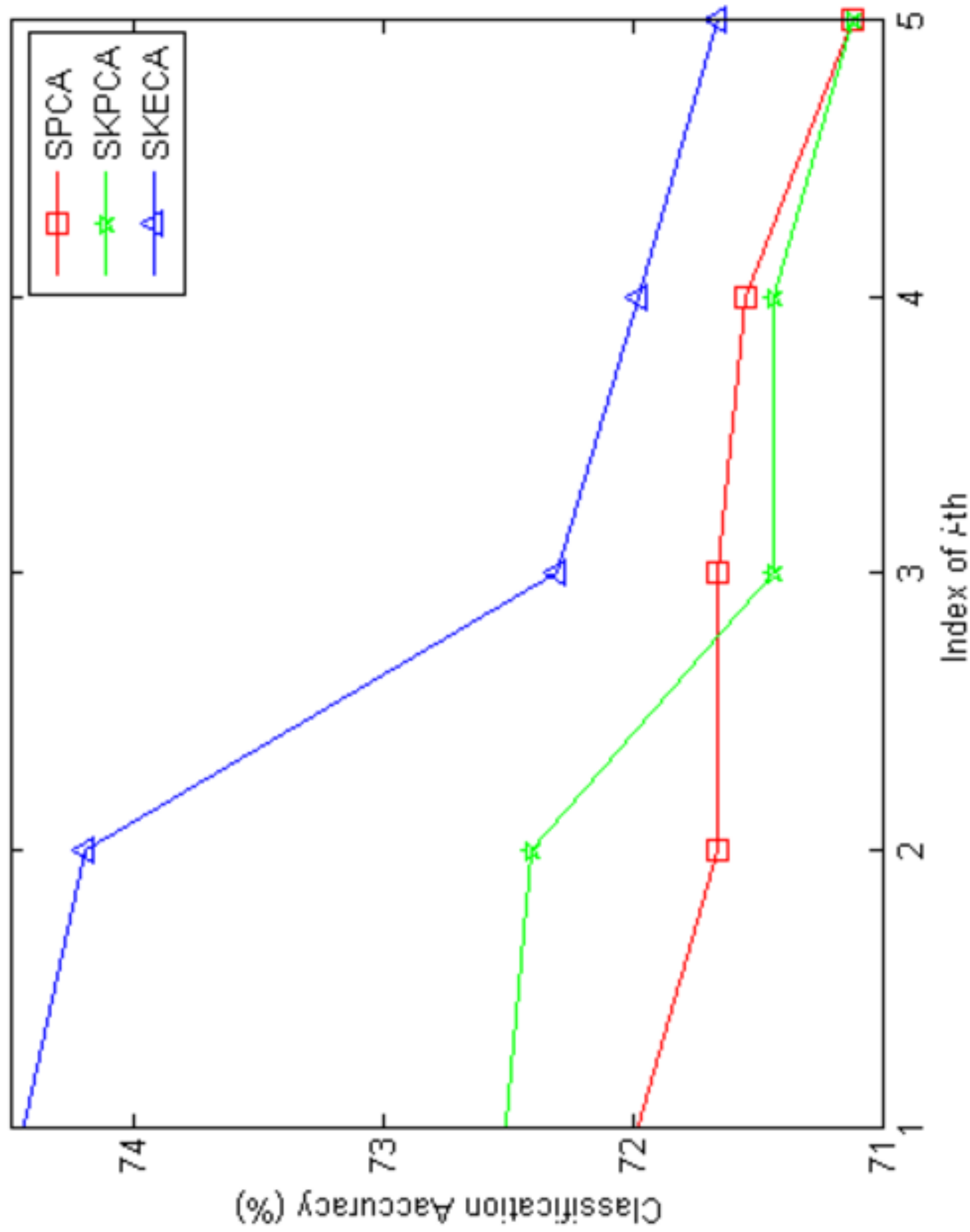
(a)


SKPCA
Sparse Parameter(t=[1,1,1,1,1])
Sparse Parameter(t=[5,5,5,5,5])
Classification Accuracy (%)
75
74
73
72
71
1
2
3
4
5
Number of Sparse Components

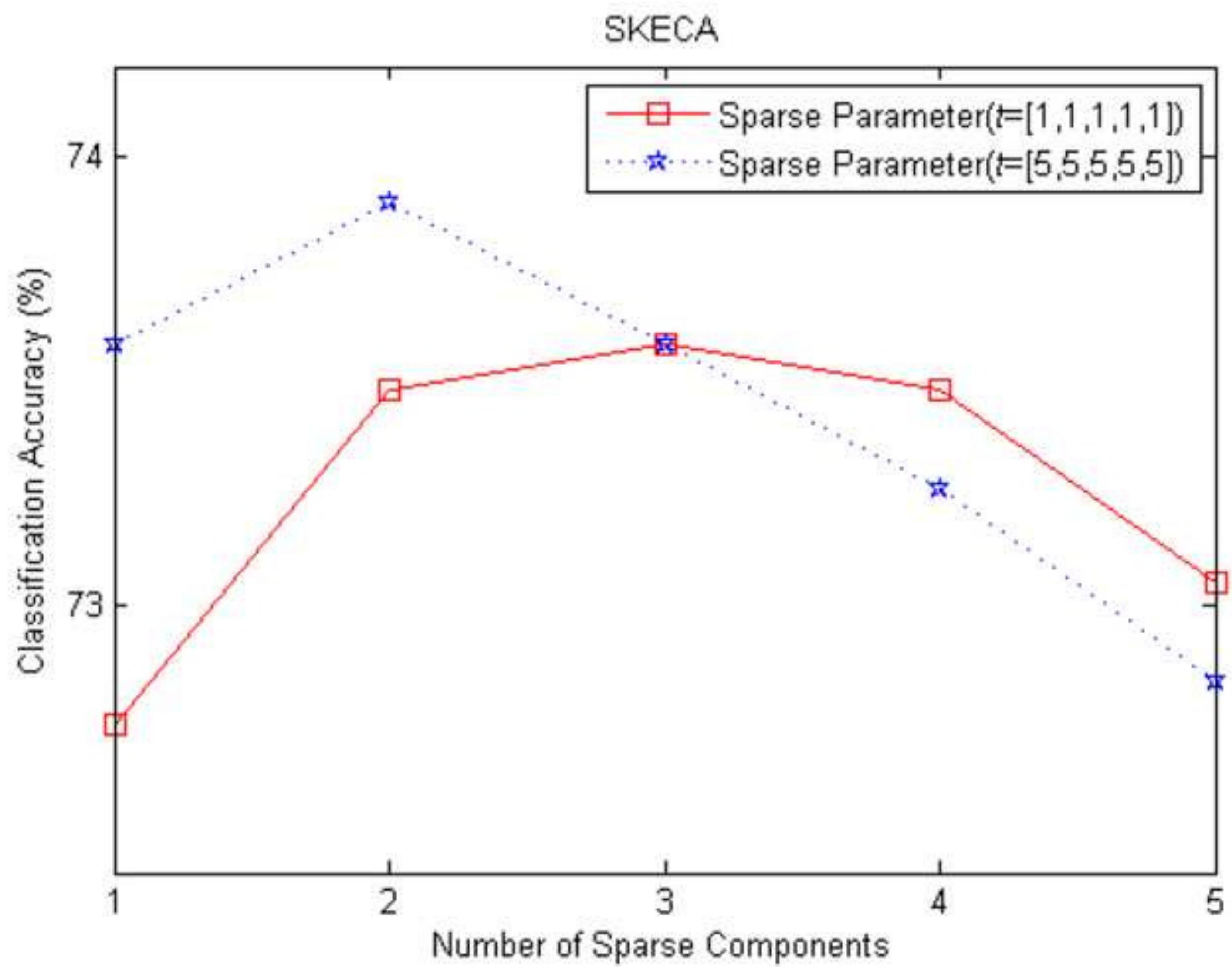
(b)


SPCA
Sparse Parameter(t=[1,1,1,1,1])
Sparse Parameter(t=[5,5,5,5,5])
Classification Accuracy (%)
Number of Sparse Components
74
72
70
68
1
2
3
4
5

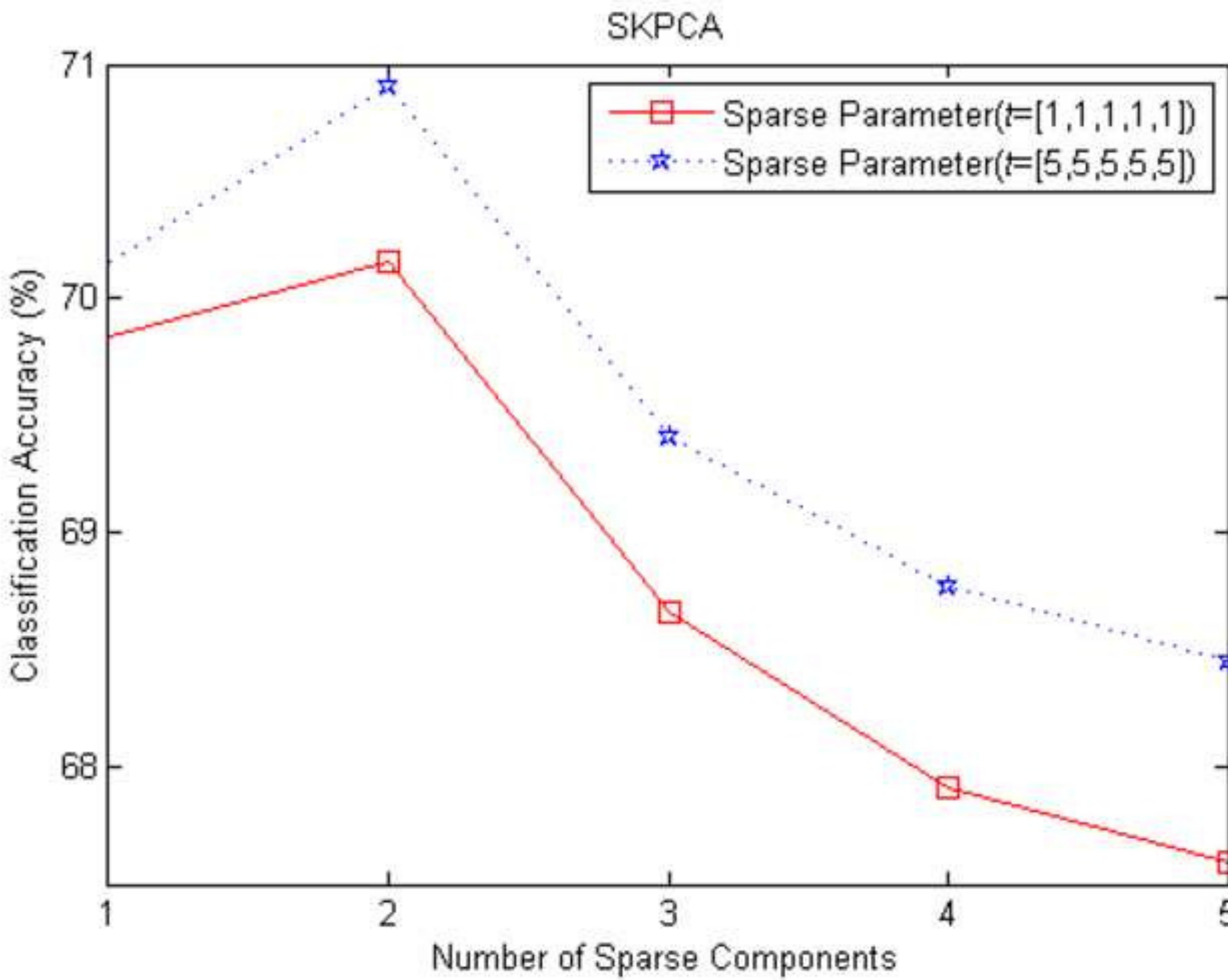
(c)



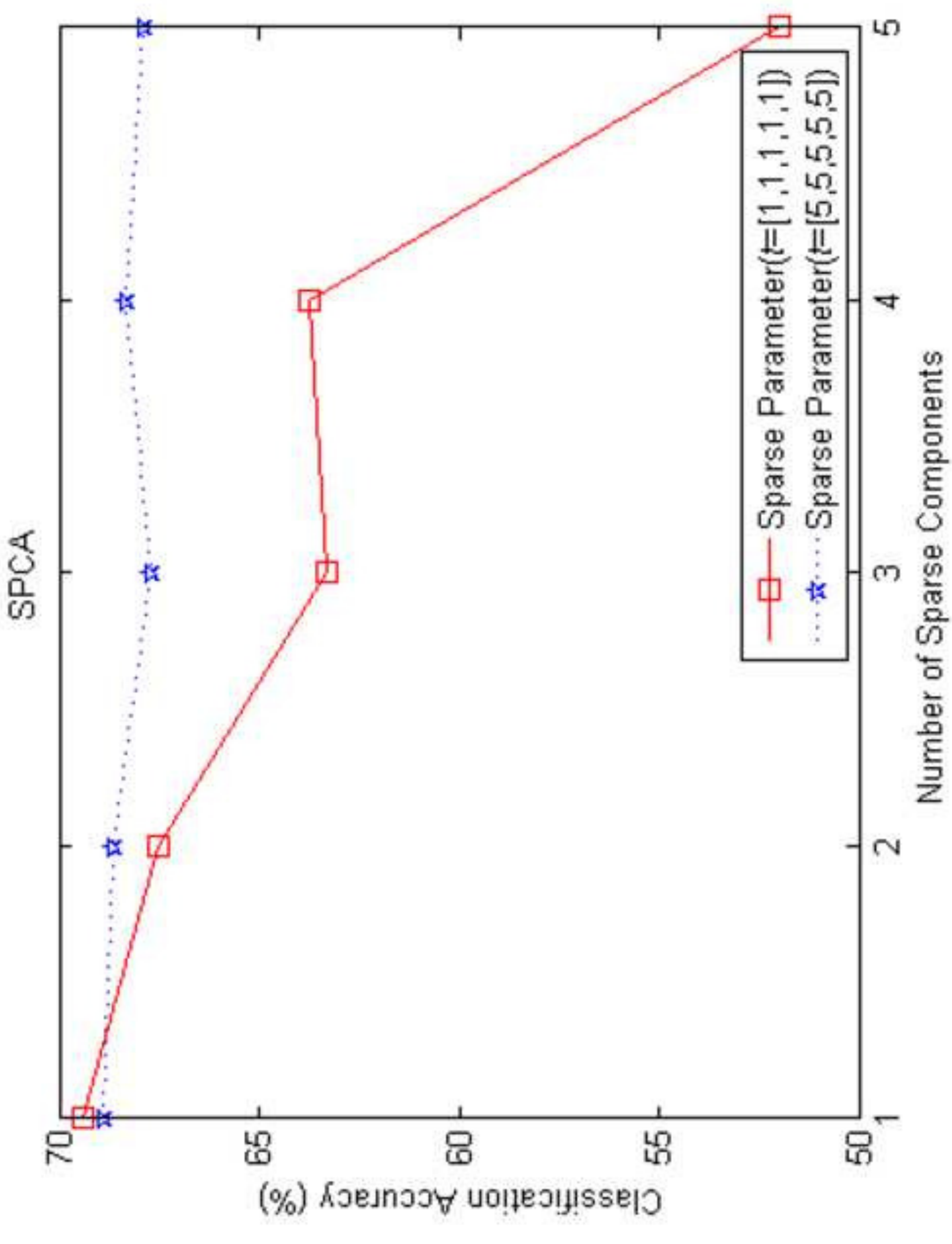
SPCA
SKPCA
SKECA
Classification Aaccuracy (%)
Index of i-th
71
72
73
74
1
2
3
4
5


SKECA
Sparse Parameter(t=[1,1,1,1,1])
Sparse Parameter(t=[5,5,5,5,5])
74
73
Classification Accuracy (%)
1
2
3
4
5
Number of Sparse Components

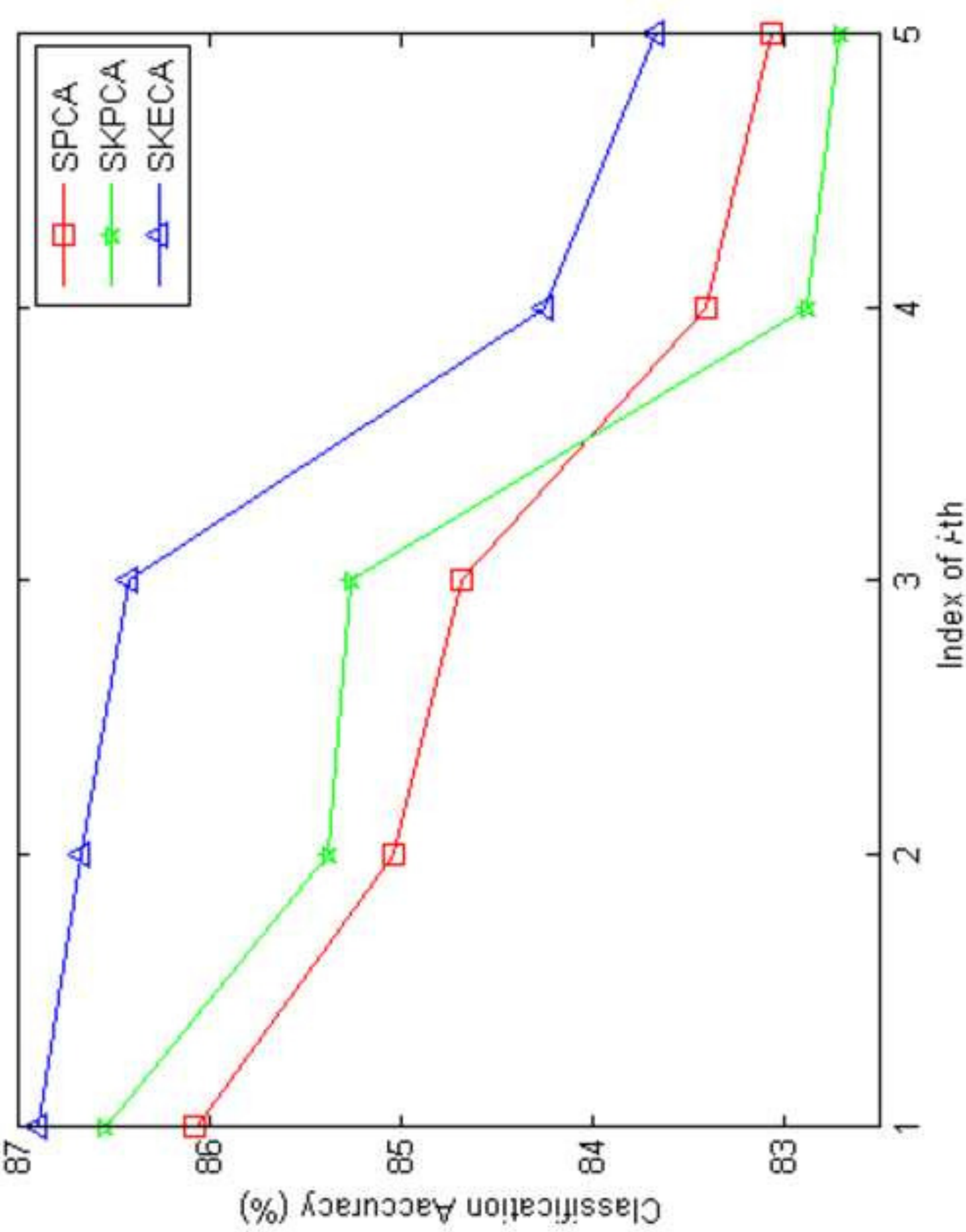
(a)

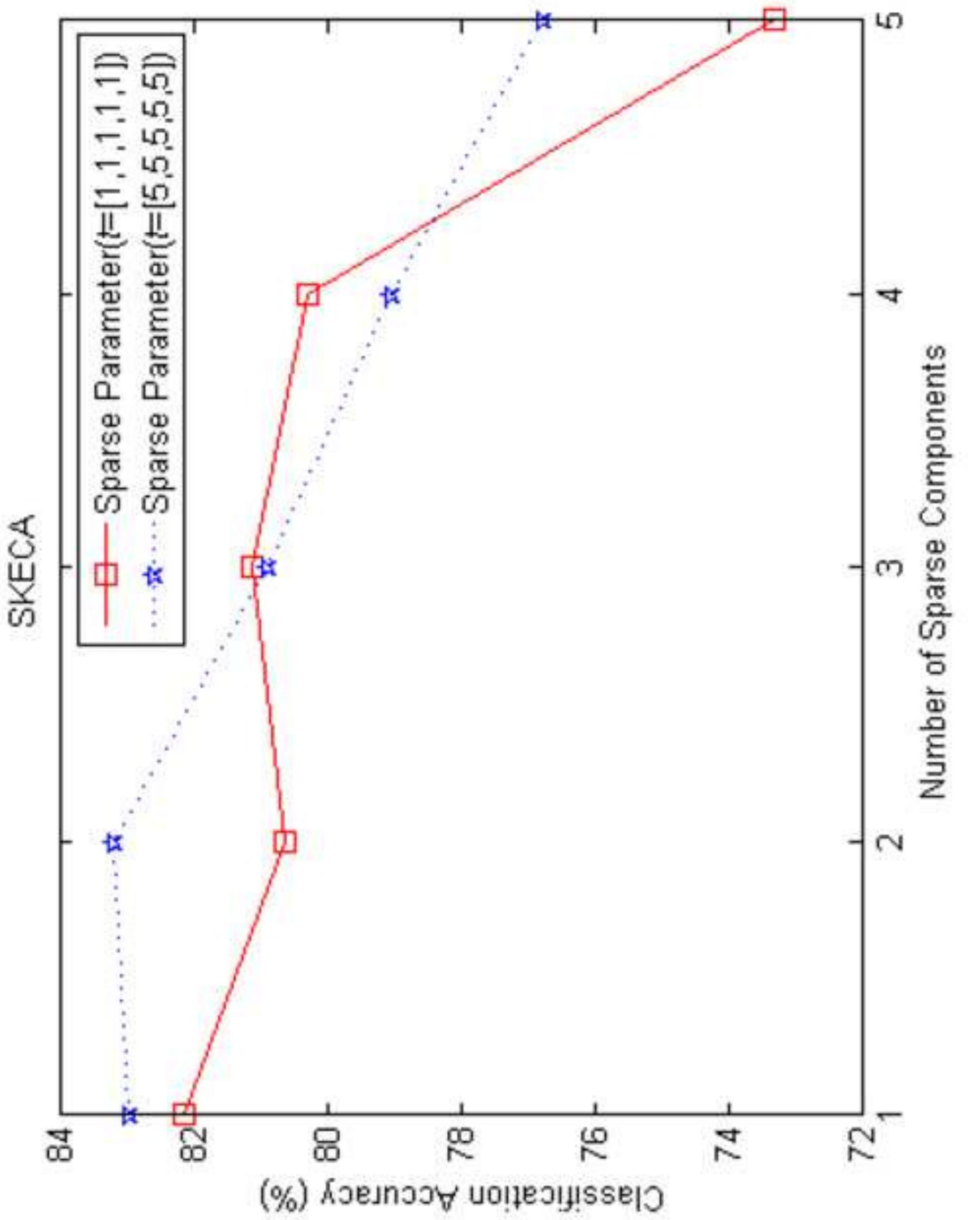

SKPCA
Sparse Parameter(t=[1,1,1,1,1])
Sparse Parameter(t=[5,5,5,5,5])
Classification Accuracy (%)
71
70
69
68
1
2
3
4
5
Number of Sparse Components

(b)

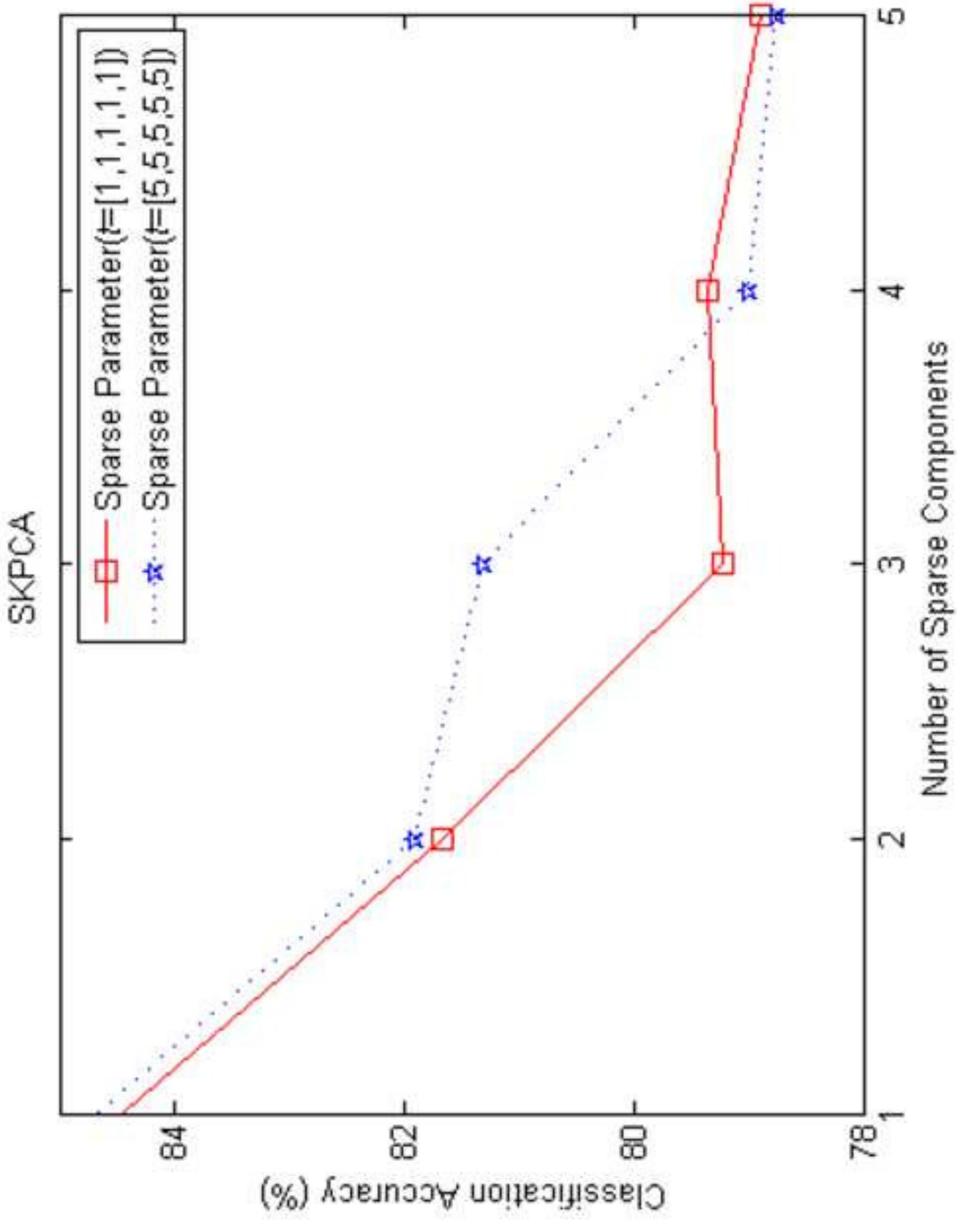

SPCA
Classification Accuracy (%)
70
65
60
55
50
1
2
3
4
5
Number of Sparse Components
Sparse Parameter(t=[1,1,1,1,1])
Sparse Parameter(t=[5,5,5,5,5])

(c)


SPCA
SKPCA
SKECA
Classification Aaccuracy (%)
Index of i-th
87
86
85
84
83
1
2
3
4
5

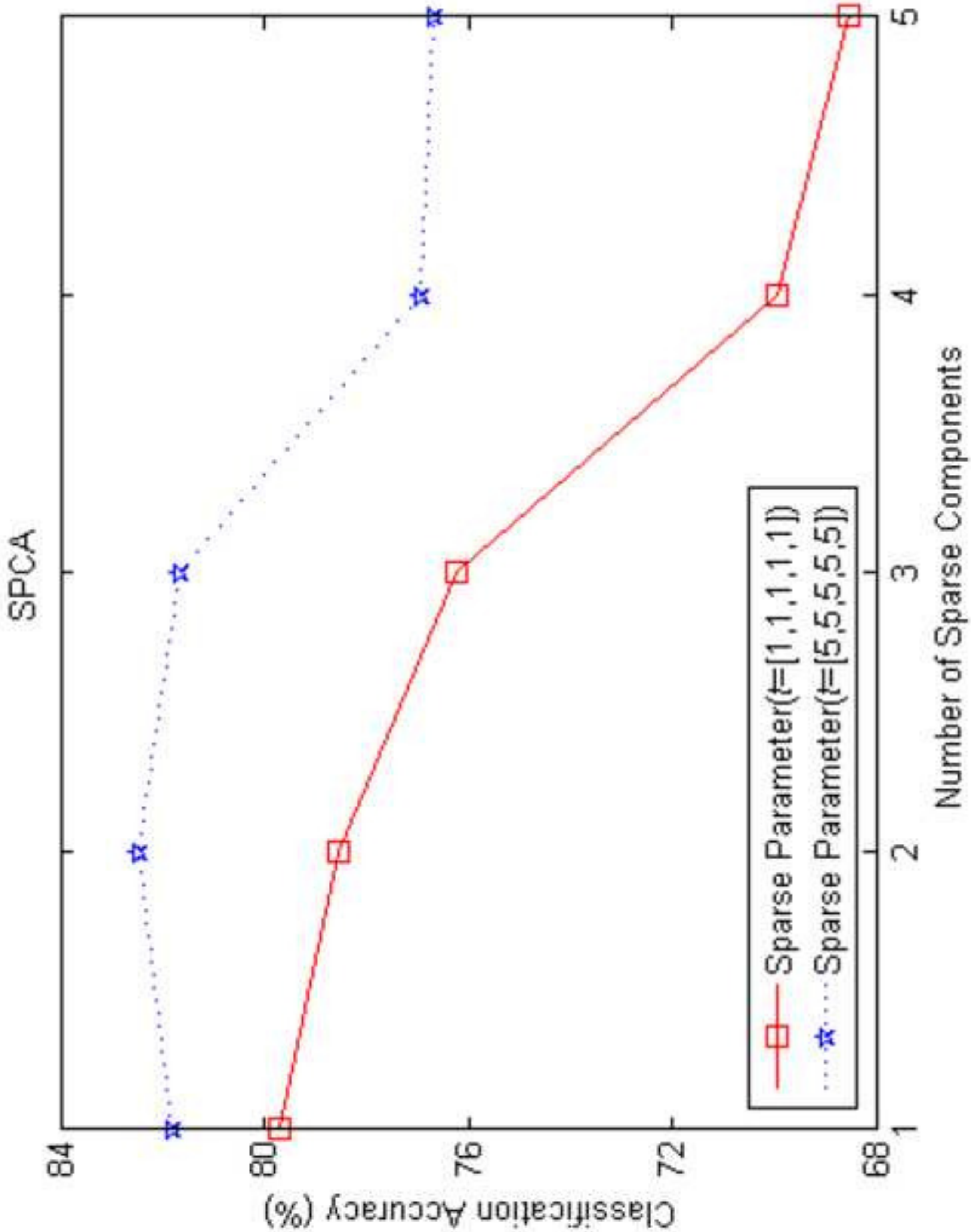

SKECA
Sparse Parameter(t=[1,1,1,1,1])
Sparse Parameter(t=[5,5,5,5,5])
Classification Accuracy (%)
Number of Sparse Components
84
82
80
78
76
74
72
1
2
3
4
5

(a)


SKPCA
Sparse Parameter(t=[1,1,1,1,1])
Sparse Parameter(t=[5,5,5,5,5])
Classification Accuracy (%)
Number of Sparse Components
78
80
82
84
1
2
3
4
5

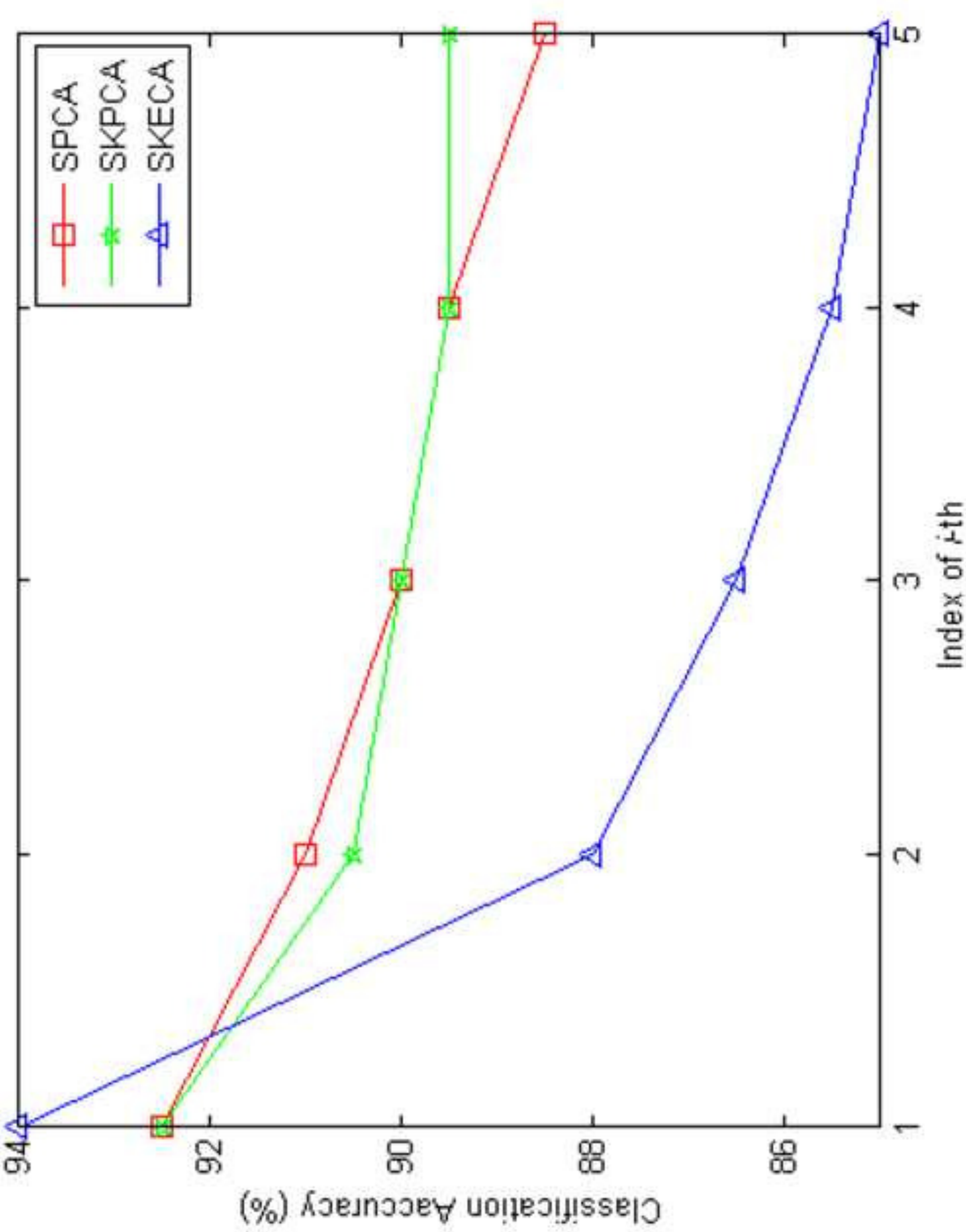
(b)

Figure


SPCA
Classification Accuracy (%)
84
80
76
72
68
Number of Sparse Components
1
2
3
4
5
Sparse Parameter(t=[1,1,1,1,1])
Sparse Parameter(t=[5,5,5,5,5])

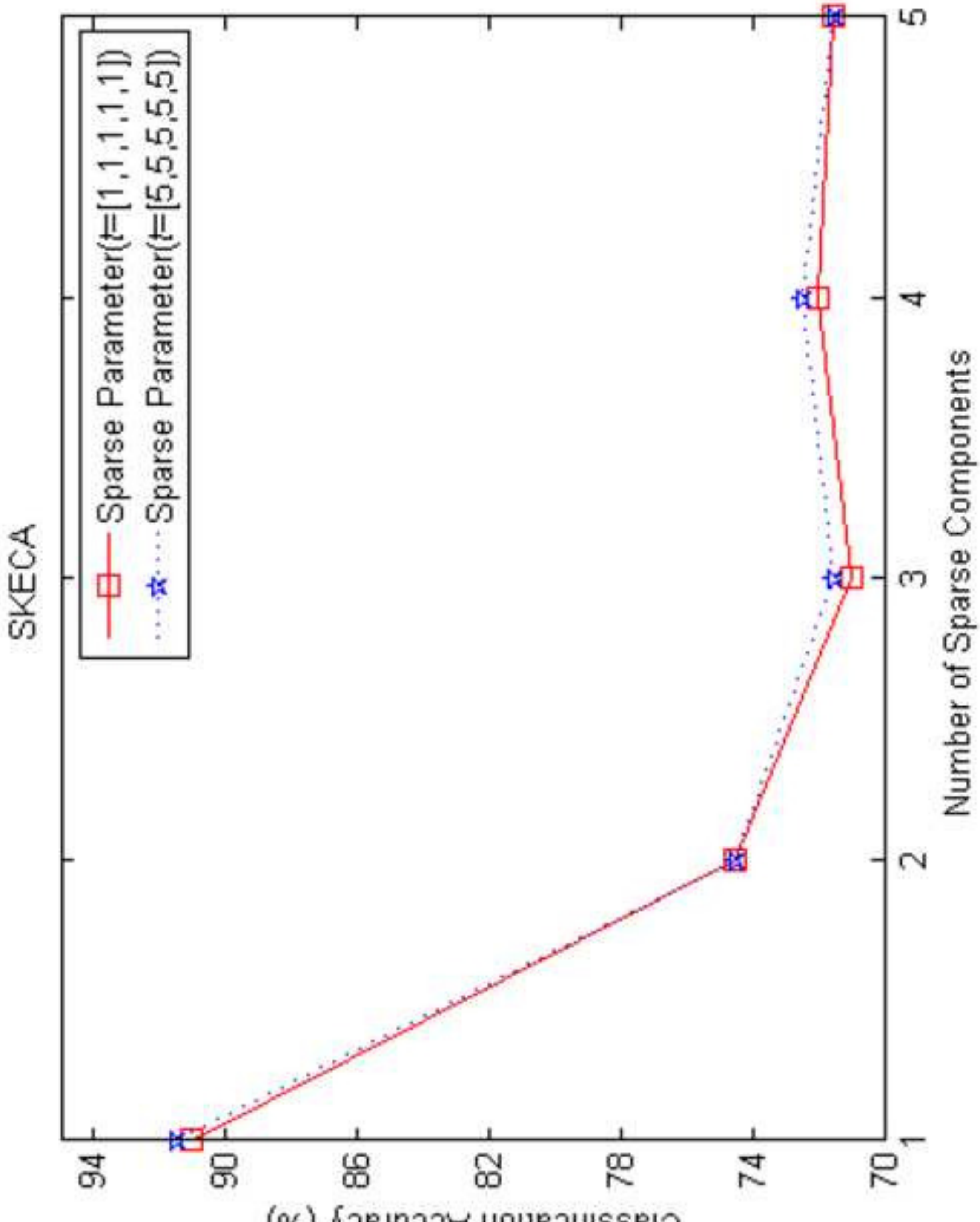
(c)

Figure

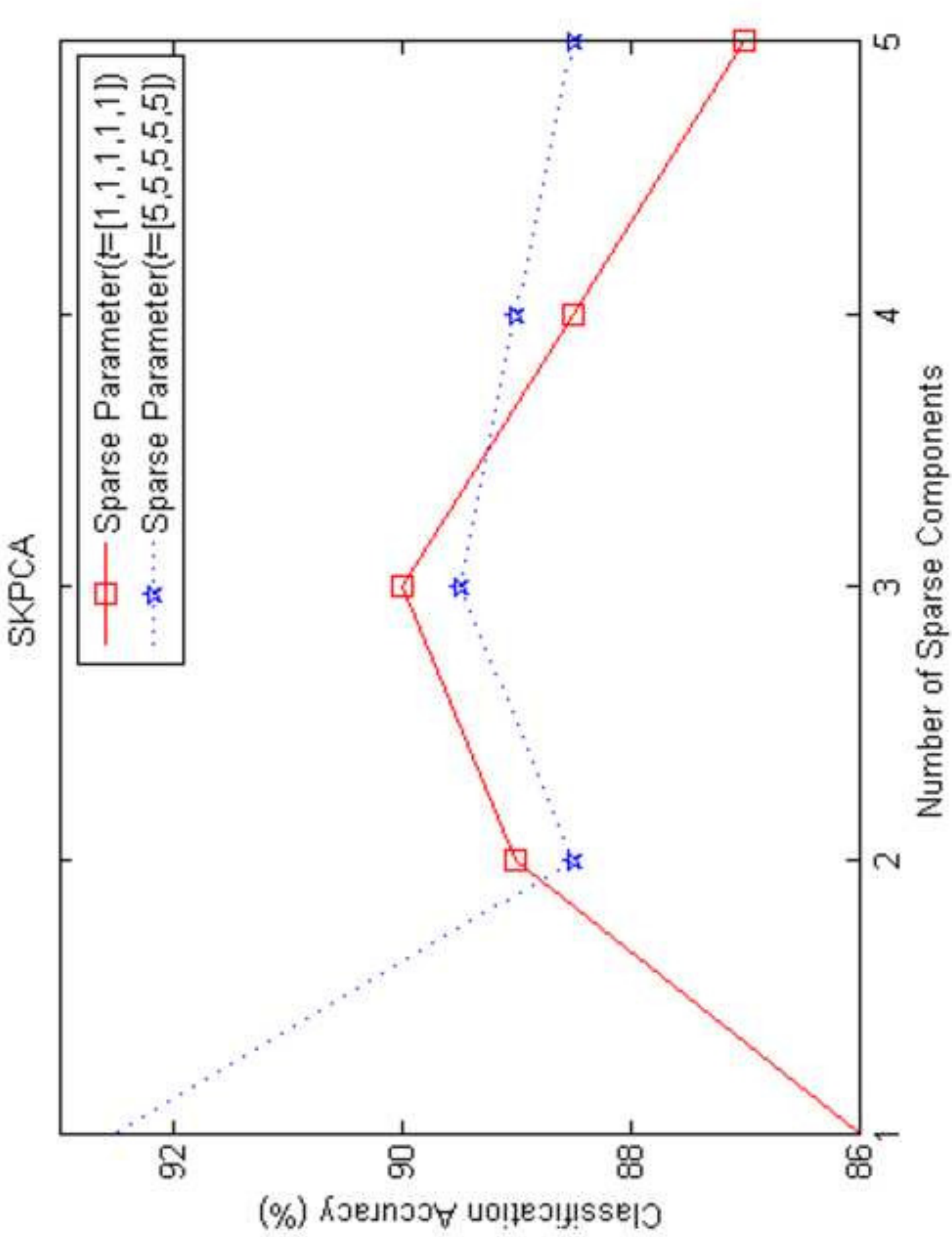

SPCA
SKPCA
SKECA
Classification Aaccuracy (%)
Index of i-th
94
92
90
88
86
1
2
3
4
5

Figure

Figure


SKECA
Sparse Parameter(t=[1,1,1,1,1])
Sparse Parameter(t=[5,5,5,5,5])
Classification Accuracy (%)
Number of Sparse Components
94
90
86
82
78
74
70
1
2
3
4
5

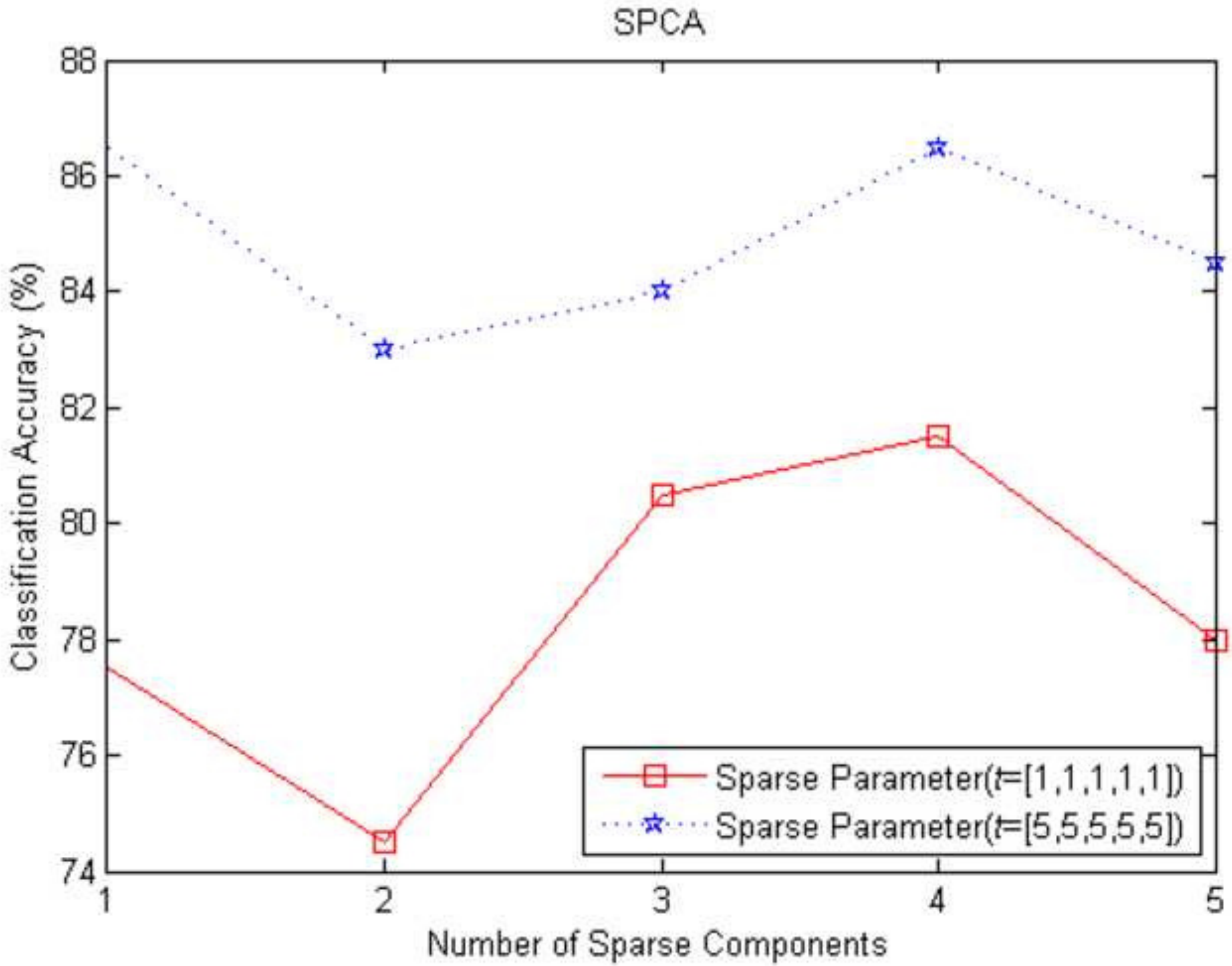
(a)


SKPCA
Sparse Parameter(t=[1,1,1,1,1])
Sparse Parameter(t=[5,5,5,5,5])
Classification Accuracy (%)
92
90
88
86
1
2
3
4
5
Number of Sparse Components

(b)

Figure

Figure


SPCA
Classification Accuracy (%)
Number of Sparse Components
Sparse Parameter(t=[1,1,1,1,1])
Sparse Parameter(t=[5,5,5,5,5])
74
76
78
80
82
84
86
88
1
2
3
4
5

(c)